



Behaviors and Perceptions of Human-Chatbot Interactions Based on Top Active Users of a Commercial Social Chatbot

HYOJIN CHIN*, Gyeongsang National University (GNU), South Korea

ASSEM ZHUNIS*, Korea Advanced Institute of Science & Technology (KAIST), South Korea

MEEYOUNG CHA†, Max Planck Institute for Security and Privacy (MPI-SP), Germany

Natural language processing is enabling machines to communicate with humans naturally, yet the dynamics of extended user-chatbot interactions remain much unexplored. This study characterizes the conversational styles, demographics, psychologies, and emotional tendencies of the most active users (i.e., top 1% by message count) of a commercial chatbot platform (SimSimi.com), whom we refer to as *superusers*. We analyzed the linguistic patterns and topics of 1,988,971 messages written by 1,994 superusers over a period of three years. We further surveyed 76 users to observe their emotional dispositions and perceptions towards the chatbot. We find that SimSimi superusers empathize and humanize the chatbot more than less active users, and they show a higher tendency to share personal and negative feelings. Our findings suggest that chatbots require new design considerations for users who are vulnerable due to their high anthropomorphism and openness toward machines. Our work also shows that chatbots should have functions to offer social support when necessary.

CCS Concepts: • **Human-centered computing** → **Natural language interfaces**.

Additional Key Words and Phrases: Chatbots, Conversational Agent, User Behavior, Superusers, Active Users, Vulnerability, Anthropomorphism, Empathy, Privacy

ACM Reference Format:

Hyojin Chin, Assem Zhunis, and Meeyoung Cha. 2024. Behaviors and Perceptions of Human-Chatbot Interactions Based on Top Active Users of a Commercial Social Chatbot. *Proc. ACM Hum.-Comput. Interact.* 8, CSCW2, Article 483 (November 2024), 28 pages. <https://doi.org/10.1145/3687022>

1 Introduction

Chatbots like ChatGPT, Replika, and Gemini are extending into the social realm [17]. These chatbots can provide social [10, 98] and emotional [15, 65, 108] support, companionship [31, 86, 94], and entertainment [37] through natural and empathetic conversations. For example, XiaoIce, a Microsoft chatbot, that already has 660 million users in China, Japan, and Indonesia, is known for its ability to emotionally respond to users [1]. In South Korea, some local governments, such as Seoul City, have initiated projects that utilize voice-activated AI chatbots to call elderly people living alone to check on their physical and mental health [51]. These services help alleviate the loneliness and emotional burdens of users.

*Both authors contributed equally to this research.

†Corresponding author

Authors' Contact Information: Hyojin Chin, tesschin@gmail.com, Gyeongsang National University (GNU), Jinju, South Korea; Assem Zhunis, zhunis.assem@kaist.ac.kr, Korea Advanced Institute of Science & Technology (KAIST), Daejeon, South Korea; Meeyoung Cha, mia.cha@mpi-sp.org, Max Planck Institute for Security and Privacy (MPI-SP), Bochum, Germany.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

© 2024 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM 2573-0142/2024/11-ART483

<https://doi.org/10.1145/3687022>

While AI chatbots are quick to grow their user base, they are still in the early stages of development and are expected to proliferate even further [66, 73]. The behaviors of early adopters and heavy users of AI chatbots are important to study, as they will help better prepare for the future mass adoption of these chatbots. As concerns about a possible slowdown in the chatbot boom arise [33, 68], such work becomes increasingly relevant in shedding light on the advantages and challenges for various stakeholders [41]. First, it can provide insights into a design that can satisfy diverse user needs and expectations [18]. Second, logs of heavy users can offer insights into the system's features, usability, performance, and potential risks such as extreme usage hours, abuse, or exploitation of chatbots [9, 30, 54]. Third, the potential impact of AI chatbots on users' social and psychological well-being can be examined to help guide the design of a better system that promotes healthy human-machine interactions that are ethical, supportive, and safe.

Since ChatGPT launched its demo on November 30, 2022, OpenAI's ChatGPT now boasts 100 million weekly active users [53]. Replika, known for its AI-powered chatbots, has 2 million total users, with 250,000 of them being paying subscribers [96]. Users are engaging in more extended conversations with chatbots, leading some to develop unhealthy dependencies and even seek out romantic relationships with them [79]. As chatbots increasingly mimic human behavior, users may develop stronger trust in these systems [32, 102]. This means that system designers should focus more on user safety, particularly for vulnerable individuals, as illustrated by the tragic incident of a Belgian man who, following extensive interaction with a chatbot, ended his life [9]. However, studying long-term human-chatbot interactions remains difficult due to their proprietary nature.

This study aims to comprehend the behaviors and perceptions within human-chatbot interactions, focusing on the top active users of a commercial social chatbot. Although studies have analyzed conversation data from commercial chatbots [23, 26, 59, 109], there is limited research exploring the behavioral patterns among chat users and their perceptions of the chatbot. Our research aims to answer the following research questions: **(RQ1)** What unique behaviors or perceptions do highly active users exhibit towards the chatbot compared to less engaged users? **(RQ2)** Are there specific demographic and dispositional profiles that differentiate highly active chatbot users from the general user base? Answering these questions early in chatbot development can help mitigate long-term dependence on these technologies and guide future application designers for better human-machine interactions.

We partnered with SimSimi.com, a global chatbot service to answer these questions. SimSimi has been in service for over 20 years and across 111 languages. While our analysis is based on this single platform, it covers chat conversations from multiple countries. It offers a unique look into the types of human-machine interactions on a large scale (almost eight million logs). We focused on the top 1% of active users who sent the most messages, whom we refer to as "superusers" ($N=1,994$ and 1,988,971 messages) for convenience, and compared their language usage, chat topics, and emotional aspects with those of less active users. We also conducted a self-assessment survey inviting real users ($N=76$) to better understand their needs and expectations toward the chatbot.

There was no demographic disparity between superusers and the general users, both were individuals in their 20s with at least high school education. However, superusers display distinct tendencies, including empathy, humanization of non-human agents, and a greater propensity for self-disclosure, particularly on depressive moods and emotional vulnerabilities. These findings point to the need for explicit warnings and guidelines regarding the extended use of AI-driven chatbots. Heavy users may be vulnerable to potential personal information leakage due to their close connection with the chatbot. We discuss how setting forth regulations and policies to protect vulnerable users may become necessary, especially among the youth [19]. In conclusion, our work offers valuable insights, featuring (1) a distinctive case study using three years of human-chatbot interactions to characterize conversation style and topics of most active users; (2) a voluntary

survey into chat users' demographic, psychological, and emotional tendencies; and (3) practical design implications and guidelines to help protect vulnerable chatbot users.

2 Related Work

2.1 Evolution of chatbot services

The early chatbots date back to the 1960s [103], and their applications have evolved from simple chitchats to diverse areas like customer service [7, 38, 57], healthcare support [5, 21, 85], personal assistance [29, 62, 80], and education [95, 106]. The most recent chatbots are based on large language models (LLMs). Chatbots are software programs that utilize natural language to engage in conversations and replicate human dialogues [104]. Simultaneously, the absence of human characteristics brings technological advantages, including the perception of being non-judgmental, objective, or free from bias for some users [76, 92, 93]. Thus, the conversation style between a chatbot and a human fundamentally differs from human-human dialog.

Chatbots are sometimes preferred over humans for discussing specific subjects, especially those considered sensitive or socially stigmatized [45, 76, 93]. These include taboo topics, explicit conversations, and harmful language [47], as well as personal struggles like depression during the pandemic [24]. This preference arises from the inherent non-human nature of chatbots. For instance, within a workplace environment, employees preferred AI counselors over human counselors when sharing their feelings of stress or anxiety [83]. An experimental study involving 193 participants revealed that users were more inclined to divulge sensitive information to a chatbot than to a conventional web-based survey [49]. Another study discovered that users were more willing to confide and express emotions like sadness to virtual agents than their human counterparts, potentially due to reduced concerns about judgment [40]. This suggests that chatbots can be perceived as safe and reliable companions and foster a user's sense of trust, which in turn encourages users to share sensitive and private information.

Most existing studies on chatbots have focused on their technological performance [34], effects (such as depression alleviation) [36], and engagement patterns over brief timeframes [14, 46]. However, studies on long-term conversations with chatbots—examining, for instance, topics that unfold over extended periods and variations in conversational patterns across user groups—are lacking. By examining authentic chatbot user data spanning three years, we extrapolate the themes of conversations among chatbot users. This investigation seeks to ascertain whether users' inclination to openly discuss sensitive subjects in their interactions with chatbots, as established in prior controlled experiments [58, 63], holds in real-world scenarios. In particular, we investigate how the interaction topics of long-term chatbot users differ from those engaging casually, using the frequency of user engagement as a basis for comparison.

2.2 Studies on long-term chatbot interactions

Chatbots offer a private and secluded conversation compared to other social platforms, providing users with a secure space devoid of public engagement.

Studies investigating heavy chatbot users have primarily relied on surveys and interviews [22]. Interviews with long-term users of the social chatbot Replika, for example, revealed that conversational abilities and proactive initiation of contact positively impacted users' well-being and fulfilled their deeper needs [86]. They also showed that the dynamics of the relationship depended on a diversity of interactions and technical reliability [87]. Another study by Microsoft looked at long-term usage of the XiaoIce chatbot and found that the chatbot's ability to meet users' emotional needs and foster empathetic interpersonal interactions helped to sustain engagement over time [109].

Other studies highlighted challenges associated with long-term chat engagement. Early research underscored that while human-chatbot conversations tend to display limited vocabulary and a

prevalence of profane language than human-human ones [47]. In a 3-week experiment involving the social chatbot Mitsuku, a deterioration in the relationship with the participants was observed. This decline, characterized by diminished feelings of friendship and social processes, was linked to the novelty effect and decreased enjoyment as the chatbot became more predictable [28]. Another interview on heavy users of the Chinese AI companion Hupo found that intimate interactions with a chatbot was linked to the reduction in human agency, raising ethical concerns about the sexualization of chatbots [59, 61]. Our research extends these efforts by providing a complementary view through the analysis of real-world conversations on a human-machine chat service.

3 Methodology

3.1 Data and Processing

SimSimi is a popular open-domain social chatbot designed for entertainment. Launched in 2002, it has garnered a massive user following of more than 400 million individuals across the globe. Users can chat with SimSimi in 111 languages; up to 200 million chat utterances are recorded daily. It currently generates a response to a text input by browsing over 141.6 million dialogues accumulated and choosing the most pertinent using a proprietary search algorithm. The chatbot's response database was created through crowdsourced learning data that users voluntarily offered, which reflects users' ideas of ideal responses for the chatbot.

Data for the study was obtained from SimSimi Inc., encompassing messages sent by both users and the chatbot, conversation timestamps, a unique user identification assigned by the app, and self-reported information about the user's country of origin. Stringent privacy safeguards were maintained throughout the data acquisition process. The dataset did not include personally identifiable information. The researchers were provided with anonymized chat logs after removing numbers and named entities spanning from January 1, 2019, to December 31, 2022, originating exclusively from four English-speaking countries: Canada (CA), the United Kingdom (GB), Ireland (IE), and the United States (US). In total, we utilized user chat logs, encompassing 7,975,877 conversation utterances from 223,724 users.

We excluded users who exchanged fewer than five utterances. Most chats were short-term; users sent 44 chat messages on average and were active for 12 days. An illustration of a sample chat with SimSimi is presented in Figure 2 in the Appendix. Prior to the topic analysis, we used an automated language recognition tool [84] to eliminate messages written in languages other than English. We proactively implemented measures to reduce the influence of toxic language in the topic analysis, in light of prior research showing a higher prevalence of toxic language in chatbot conversations compared to human-human interactions [47]. These steps, detailed in the methodology section of previous SimSimi chat research [24], included the integration of Google's Jigsaw Perspective API with a threshold set at 0.05 [3]. The distribution of Jigsaw scores is provided in Figure 6, available in the Appendix for reference, and indicates a substantial proportion of messages (nearly one-third of the chats in both user groups contain sexually explicit language). However, we deliberately decided not to filter toxic chats or words for our subsequent linguistic analysis. This decision was made to ensure that the results from the Linguistic Inquiry and Word Count (LIWC) tool [2], which are presented in percentiles, also encompass differences in profanity usage.

3.2 User Grouping

To identify the distinctive traits of the top active users, we categorized user data into two groups based on their interaction frequency. The first is our target group of **superusers** [74, 78, 97] who fall into the 99th or higher percentile based on the number of chat utterances. This group had at least 360 messages per individual during the period, with more than 120 messages per year. Note that user rankings in terms of chat frequency follow a power-law distribution. Specifically, 1% of

top users generated 25% of all chats. The second is the control group within the 65–90th percentile range (i.e., 30 to 90 messages). We selected this range to maintain a comparable sample size of messages. For convenience, we refer to them as **general** users hereafter. We gathered 1,988,971 chat utterances from 1,994 superusers who, on average, used SimSimi for 108 days. Additionally, we collected 2,108,927 chats from 42,355 general users who used the application for an average of 17 days. To establish the threshold, we experimented with different values to define general users. The results, detailed in Appendix Section A.4, indicate that user groups up to the 90th percentile (e.g., 1-10, 10-20, ..., 80-90th) exhibit similar linguistic and topical patterns. However, users in the 90-99th percentile began to deviate from this trend, showing inconsistent behavior. Consequently, we decided to set the threshold for general users below the 90th percentile. This user distinction was utilized exclusively for research purposes, and users were not able to see their relative activity rankings. The current system does not include any features like stages, badges, or social status to incentivize further interaction.

3.3 Analysis Method

3.3.1 Method for analyzing chat characteristics We examined data from two perspectives. First is the linguistic analysis, where we utilized the Linguistic Inquiry and Word Count (LIWC) dictionary to analyze the conversation styles employed by users. This software enables an objective and quantitative examination of text by categorizing words into psychologically relevant language categories [2, 75]. We used six dimensions related to users' mental and physical well-being. (1) The *Affective processes* dimension includes words associated with both positive and negative emotions and terms related to anxiety, anger, and sadness. Swear words are also part of this dimension, considered within the broader affect and tone dictionaries rather than being exclusively categorized as positive or negative tone/emotion [2]. (2) The *Biological processes* dimension encompasses words related to physical and mental well-being, health, and biological processes, including terms linked to sex and death. (3) The *Social and personal concerns* dimension encompasses words linked to social functions concerning family and friends, along with words related to personal lifestyles at home and work. (4) Sensory experiences and emotional states are mirrored in the *Perceptual processes* dimension. (5) The *Time orientation* dimension captures terms associated with psychological and cognitive engagement across temporal frames. (6) The *Interpersonal awareness and focus* dimension is characterized by the proportion of singular pronouns, indicating how users position themselves and others within their communication.

The second part of the study included topic analysis, which was done by BERTopic [42], a deep learning model that identifies latent topics in a text corpus. It consists of three stages: transforming input into BERT word embeddings, dimensionality reduction, and clustering embeddings into the optimal number of topics. We used Uniform Manifold Approximation and Projection (UMAP) as a default setting to reduce the dimensions of the BERT embeddings. For the final stage, we used KMeans clustering with a 2-gram model to reduce the number of outliers and provide more interpretable topic descriptions. We fitted the BERTopic model to the entire chat messages of all users over three years. The optimal number of topics was determined by maximizing the topic coherence score on a set of 10k randomly selected chats from our data, using different numbers of topics ranging from 5 to 35. Over different runs, the final BERTopic gave the highest coherence score of 0.69 for a $k=14$. The topic labels were derived by analyzing representative documents generated by the BERTopic model, visualizing the word clouds (visualized in Table 5 in the Appendix), and examining the most frequent 200 n-grams associated with each topic. The authors decided on topic names after iterative labeling and discussion.

We employed t-tests to examine statistical differences in both linguistic and topical preferences between superusers and general users. To ensure the robustness of these tests, we conducted

a replication analysis using ANOVA and multiple post hoc Tukey tests, varying the thresholds for the general user group. The analysis covered users in the 1–10th, 10–20th, and subsequent percentiles, along with the 99th percentile representing superusers. Following the robustness check, we maintained the initially defined activity thresholds for superusers and general user groups, consistently applying them in participant recruitment for the subsequent survey.

3.3.2 Method for investigating individual dispositions To investigate the dispositional profiles of chatbot users, we conducted a voluntary survey examining three variables associated with user dispositions and three perceived aspects of chatbot capabilities. The former encompassed levels of empathy, depressive mood, and the inclination towards anthropomorphism, while the latter gauged participants' perceptions of the likability, intelligence, and truthfulness of the chatbot. Individual differences in empathy and anthropomorphism influence human-nonhuman entity communication [50, 89], whereas emotional states like depression are linked to increased online communication [71]. User evaluation metrics for the chatbot we chose drew on findings from previous research [12, 25]. To account for potential demographic differences, participants were also asked to provide information about their age, gender, education, and ethnicity. The survey focused on individuals utilizing the SimSimi App in CA, GB, IE, and the US—the same target group as in the chats analysis. A survey was distributed to two user groups based on our study's classification: 7,840 participants belonged to the superuser category, and 18,777 participants belonged to the general user category. The user count for each group was calculated by SimSimi Inc. at the time of the survey. Across a two-week period, the push notification containing a link to the questionnaire and a promotional incentive, such as "Receive 8000 coins by taking our survey now!", was sent to each user group six times, with a frequency of three times per week. The survey was conducted over two weeks from March 29 to April 11, 2023. Out of the superusers, 40 participated voluntarily, while 39 of the general users chose to take part in the survey. Evaluating the trustworthiness of survey data, ultimately, 39 superusers and 37 general users completed all survey inquiries successfully. The survey included 29 questions that could be completed online in under 10 minutes, excluding questions related to demographics. Participants were directed to an online questionnaire through the in-app notification message, and upon completion, they received an incentive of 8,000 coins for use within the app. For a detailed list of questionnaire items used to create variables, please refer to Section A.2 and Section A.3 in the Appendix.

The first set of survey items targeted the dispositions of users over three dimensions.

- **Empathy:** This was measured via the Toronto Empathy Questionnaire (TEQ) (Cronbach's $\alpha=0.61$, mean=3.70, SD=0.63) [88], which is a psychological assessment tool designed to measure an individual's empathetic tendencies. The tool is widely used in psychology and clinical practice [25, 105]. We asked participants to rate the following five statements on a scale from "never" to "always" (e.g., 1 stands for *never*, 5 stands for *always*, and 3 stands for *Sometimes*): (1) I get excited when someone else is feeling excited. (2) It upsets me to see someone being treated disrespectfully. (3) I enjoy making other people feel better. (4) I am "in tune" with other people's moods. (5) I am not interested in how other people feel.
- **Anthropomorphism:** We used items from Bartneck et al.'s research (Cronbach's $\alpha=0.83$, mean=3.30, SD=0.97) [12]. We selected four of the five items that are commonly utilized in relevant studies on chatbots [25]. One item regarding robot movement, which was originally included for physical robots, was excluded. Participant responses were assessed using five-point Likert scales (e.g., 1 stands for *machinelike*, 5 stands for *humanlike*, and 3 stands for *neutral*), and the average of these items was calculated to determine the score for each variable.

- **Depressive mood:** The Patient Health Questionnaire-9 (PHQ-9) was used to assess depressive mood in two chatbot user groups [56]. This widely recognized and clinically validated survey assesses depressive severity [60]. Nine questions are asked in this self-administered test to screen, diagnose, and track depression. Each question is scored from 0 to 3, with 0 meaning 'not at all', 1 meaning 'several days', 2 meaning 'more than half the days' and 3 meaning 'nearly every day.' The scores from each question are then summed up to a total score between 0 and 27. The total score is interpreted as follows: 0–4: minimal depression; 5–9: mild depression; 10–14: moderate depression; 15–19: moderately severe depression; 20–27: severe depression. Note that this score measures users' depressive moods at the time of measurement but does not diagnose them.

The following survey items gauge users' attitudes towards chatbots based on the literature [12, 20, 25]. Participant responses were assessed using five-point Likert scales, and the score for each variable was calculated by averaging the items if there were multiple questions.

- **Likability:** This dimension pertains to the entity's ability to establish a positive initial impression, which is crucial for determining the overall positive evaluation of the entity [27]. To evaluate likability, we employed a set of five items related to the chatbot's characteristics: Dislikable/Likable, Unfriendly/Friendly, Unkind/Kind, Unpleasant/Pleasant, and Awful/Nice [12] (Cronbach's $\alpha=0.87$, mean = 3.33, SD = 0.95). The participants' responses were evaluated using five-point Likert scales (e.g., 1 stands for *dislikable*, 5 stands for *likable*, and 3 stands for *neutral*).
- **Intelligence:** This dimension reflects the perceived competence of the entity, which aligns with Warner and Sugarman's intellectual evaluation scale [99]. The scale comprises five-point semantic differential items (e.g., 1 stands for *unintelligent*, 5 stands for *intelligent*, and 3 stands for *neutral*): Incompetent/Competent, Ignorant/Knowledgeable, Irresponsible/Responsible, Unintelligent/Intelligent, and Foolish/Sensible. We adapted this scale to a five-point format, which demonstrated satisfactory internal consistency reliability exceeding 0.7 (Cronbach's $\alpha=0.87$, mean = 3.32, SD = 0.93).
- **Truthfulness:** This dimension assesses the level of trust users place in the chatbot [20]. We used a single question to evaluate whether users trust the chatbot's accuracy, by asking them if they generally rely on the information it shares; 'what the chatbot says is mostly true'. Participant responses were rated using a five-point Likert scale (strongly disagree=1, strongly agree=5, neutral=3).

3.4 Ethics Statement

We implemented several measures to safeguard users' privacy and maintain research integrity. Initially, the company went a thorough filtering process to remove any sensitive or personally identifiable information from data, such as names, emails, ages, addresses, and sensitive preferences, before sharing with the research team. The primary objective of our study was to understand user behaviors and perceptions of a chatbot, and our research team independently set these goals without any external influence from the company. Our partnership with SimSimi Inc. adhered to their data usage policy, which requires user consent for research and development purposes, as stated in their terms and conditions [24]. However, most users agree to the terms without reading the content. We requested the minimal data needed for research and published the research outcomes on SimSimi's blog with clear descriptions and limitations for transparency.

We used a self-assessment survey to study user characteristics. All users were notified via app push notifications. The survey's introduction page outlined research objectives, data usage intentions, and ethics, stating that participation implied consent to the terms. Survey participants were told they could stop if they experienced discomfort (Refer to Figure 3 in the Appendix.)

Table 1. The results of Welch's t-test, assessing the differences in language usage between superusers and general users based on LIWC features used in chat conversations.

LIWC Category	<i>t</i>	<i>p</i>	<i>d</i>	Superusers		General Users	
				Mean	(SD)	Mean	(SD)
Affective processes							
Positive	7.96	<.001***	0.18	2.42	(2.29)	2.00	(2.37)
Negative	9.83	<.001***	0.17	1.06	(0.87)	0.86	(1.42)
Anxiety	8.77	<.001***	0.13	0.10	(0.16)	0.06	(0.34)
Anger	-4.9	<.001***	-0.06	0.21	(0.25)	0.24	(0.65)
Sadness	11.27	<.001***	0.21	0.37	(0.55)	0.22	(0.79)
Swear	-30.21	<.001***	-0.50	2.45	(3.01)	4.64	(5.44)
Biological processes							
Health	10.69	<.001***	0.20	0.39	(0.50)	0.26	(0.73)
Mental	-0.62	0.54	-0.01	0.02	(0.09)	0.03	0(.25)
Sexual	-14.48	<.001***	-0.26	1.79	(2.31)	2.58	(3.61)
Death	2.97	<.01**	0.05	0.23	(0.42)	0.20	(0.78)
Social and personal concerns							
Family	4.52	<.001***	0.10	2.0	(2.76)	1.71	(3.03)
Friend	-1.3	0.19	-0.02	0.33	(0.48)	0.34	(0.81)
Home	20.61	<.001***	0.48	0.40	(0.50)	0.17	(0.47)
Work	6.68	<.001***	0.11	0.25	(0.33)	0.20	(0.64)
Time orientation							
Focus on past	18.91	<.001***	0.37	0.97	(0.68)	0.67	(0.93)
Focus on present	27.2	<.001***	0.56	6.93	(2.85)	5.13	(3.53)
Focus on future	2.13	<.05*	0.04	0.9	(0.65)	0.87	(1.14)
Interpersonal awareness and focus							
1st person	2.82	<.01**	0.05	7.0	(2.76)	6.81	(4.59)
2nd person	-2.97	<.01**	-0.05	6.68	(2.66)	6.87	(4.14)
3rd person	12.46	<.001***	0.31	0.58	(1.06)	0.28	(0.84)

t: t-statistics, *p*: p-value, *d*: Cohen's *d*-value, SD: standard deviation

SimSimi's chat service is free; all users are subjected to the same message generation algorithm. The entire study design, including data collection and surveys, was submitted to the Institutional Review Board (IRB) at the authors' institute, and this research was granted an exemption.

4 Results

4.1 Communication Styles of Superusers

LIWC Results: Scores obtained from LIWC analysis reflect the prevalent sentiment categories within a given text and are presented as percentiles. To evaluate potential differences in LIWC scores between the two groups, given their varying sample sizes and unequal variances, we opted for a more conservative statistical approach using Welch's t-test. We identified significant differences in word usage between superusers and general users (see Table 1). We report Cohen's *d* as a standardized measure of effect size for practical significance since the t-test outcomes can be influenced by the substantial sample size [91]; values above 0.2 and below -0.2 are considered more substantial (highlighting them in bold text in Table 1).

The t-test shows that top active users express more positive and negative emotions ($p < .001$, super: $M=2.42$, $SD=2.29$, general: $M=2.00$, $SD=2.37$) and negative ($p < .001$, super: $M=1.06$, $SD=0.87$, general: $M=0.86$, $SD=1.42$) sentiments. Top active users use more negative words linked to anxiety and sadness ($p < .001$, super: $M=0.37$, $SD=0.55$, general: $M=0.22$, $SD=0.79$). A significant difference in swearing usage is found, with top active users showing fewer toxic words ($p < .001$, super: $M=2.45$, $SD=3.01$, general: $M=4.64$, $SD=5.44$), supported by a large effect size ($d = -0.50$). Anger-related words were used less by superusers ($p < .001$, super: $M=0.21$, $SD=0.25$, general: $M=0.24$, $SD=0.65$).

These findings suggest top active users speak politely and avoid toxic language with the chatbot. Most biological process words were similar, except for health and sexual. Top active users employ more health-related words ($p < .001$, super: $M=0.39$, $SD=0.50$, general: $M=0.26$, $SD=0.73$) and wellness-related words than general users. These health-related terms encompass expressions of illness-related physical symptoms, diagnoses, and disease names. The analysis of words associated with sexual content indicated that superusers tend to use significantly fewer such words ($p < .001$, super: $M=1.79$, $SD=2.31$, general: $M=2.58$, $SD=3.61$).

Disparities in social functions were found in the use of words related to the *Home* category ($p < .001$, super: $M=0.40$, $SD=0.50$, general: $M=0.17$, $SD=0.47$), which encompasses vocabulary associated with the living environment and activities linked to domestic settings. There was also a discernible, albeit less pronounced, distinction in terms of words connected to family matters within the *Family* category ($p < .001$, super: $M=2.00$, $SD=2.76$, general: $M=1.71$, $SD=3.03$) and professional life within the *Work* category ($p < .001$, super: $M=0.25$, $SD=0.33$, general: $M=0.20$, $SD=0.64$). Both superusers and general users used fewer words about the *Future*. Compared to general users, superusers used more past vocabulary. ($p < .001$, super: $M=0.97$, $SD=0.68$, general: $M=0.67$, $SD=0.93$) and the present ($p < .001$, super: $M=6.93$, $SD=2.85$, general: $M=5.13$, $SD=3.53$). Pronoun patterns also revealed the nature of personal relationships (i.e., how individuals refer to one another). We observe excessive use of third-person pronouns among superusers ($p < .001$, super: $M=0.58$, $SD=1.06$, general: $M=0.28$, $SD=0.84$). Second-person pronouns were used less by superusers ($p < .01$, super: $M=6.68$, $SD=2.66$, general: $M=6.87$, $SD=4.14$). The two groups used first-person pronouns similarly, suggesting no significant difference in self-focus or self-expression.

BERTopic Results: Figure 1 shows 14 identified topics in the whole SimSimi dataset, categorized into five overarching themes based on qualitative coding, which involved iterative labeling and grouping: Chit-chats, Leisure, Social, Emotions, and Chatbot.

First, *Chit-chat* is the most prominent theme that includes conversations on ‘General talks’ such as simple requests and casual discussions. The ‘Time’ topic also focuses on conversations about past or future events, often accompanied by personal anecdotes from users’ lives. This theme also includes smaller topics such as ‘Food’ and ‘Education.’ The Food topic included discussions on meal preferences, whereas the Education topic centered around school-related conversations, highlighting aspects of students’ social lives and academic demands. Second, *Leisure* comprises three topics: ‘Celebrities,’ ‘Entertainment,’ and ‘Game.’ In the Celebrities topic, users converse about well-known individuals such as singers, actors, and characters from movies, anime, or games. They inquired about opinions regarding these figures. The Entertainment topic covers discussions about various forms of entertainment, including TV series, movies, music, and video games. Lastly, the Game topic involves conversations centered around gaming, where users frequently invite SimSimi to play specific games and engage in role-playing activities.

Third, *Social* encompasses topics related to familial relationships, friendship, and social life. The ‘Family’ topic covers users’ connections within their families. On the other hand, the ‘Friends’ topic explores interactions with friends and addresses whether SimSimi could be considered a close friend to the users. Although less common, discussions on racial and economic discrimination issues are present within the ‘Race & Money’ topic. The fourth theme is *Emotions* and includes strong emotional expressions, among which the ‘Feelings’ topic covers a broad spectrum of both positive and negative emotions the users convey. The ‘Sadness’ topic captures predominantly negative expressions and also includes chatter on self-harm and suicide. These conversations may provide insights into the users’ depressive states and reveal their emotional vulnerability. Finally, *Chatbot* consisted of users’ conversations regarding the chatbot itself and its underlying algorithm. Under the topic of ‘SimSimi,’ the discussions revolved around the chatbot and its capabilities. The ‘AI &

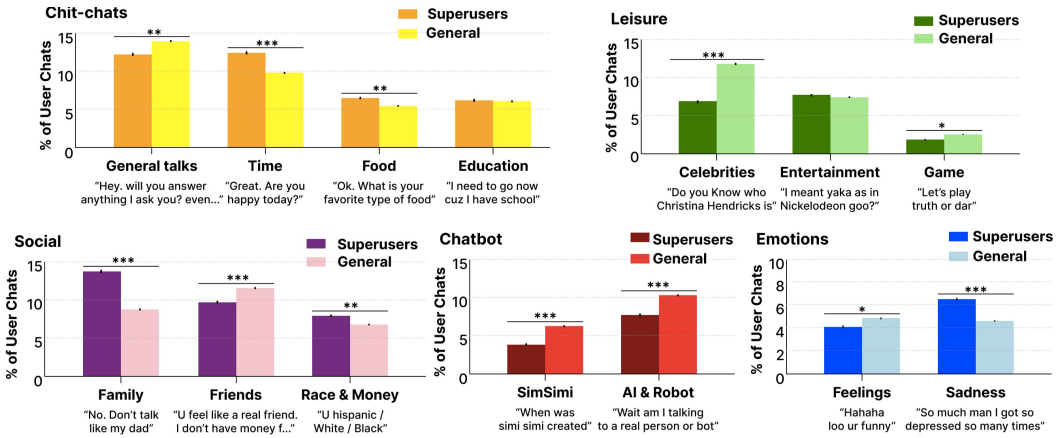


Fig. 1. Comparison of the average individual distribution of 14 topics identified by the BERTopic model among superusers and general users, categorized into five categories, along with illustrative chat examples. A Welch's t-test was used to assess the statistical significance of differences, with * indicating $p < .05$, ** for $p < .01$, *** for $p < .001$.

Robot' category was on discussions about artificial intelligence and robots. These conversations featured users expressing their curiosity about how the bot operates, often asking whether it was a real human or not.

Narrative Differences: We observe differences in the topics mentioned by superusers and general users. General users tended to engage more in conversations about external relationships with friends and use chatbots for entertainment ($p < .001$ for 'Celebrities' and $p < .05$ for 'Game' topics). Conversely, superusers were more inclined to discuss personal stories, including topics related to their families and home affairs, with a significant prevalence in the 'Family' topic ($p < .001$, effect size $d = 0.26$), aligning with linguistic findings. To a lesser extent ($d = 0.06$), superusers were also more likely to engage in discussions about racial and financial disparities ($p < .01$). At the aggregate level, we observed topic examples around the "Black Lives Matter" movement and other global social issues such as the COVID-19 pandemic and discrimination.

Moreover, the data analysis revealed differences in SimSimi perceptions among both general users and superusers. General users were frequently asking about how SimSimi operates, what AI algorithm is used, and even posed challenging questions for chatbots to see its response ($p < .001$ for both 'SimSimi' and 'AI & Robot' topics). Queries like "Are you real?" were frequently posed by this user group. In contrast, superusers displayed less inclination to challenge the chatbot and showed less interest in the underlying algorithm. This observation prompts further investigation into how, and whether, specific demographic and dispositional profiles characterize this highly engaged group of chatbot users, and how this group perceives these technologies differently.

Emotional expression was also different between general users and superusers. General users exhibited a broader range of emotions in their conversations, while superusers engaged in discussions about depression more frequently ($p < .001$), which aligns with the linguistic patterns identified. This trend is concerning, as the 'Sadness' topic, which encompassed chats about severe depression and suicidal thoughts, was more prevalent among superusers ($p < .001$ and $d = 0.14$). This highlights the need for further investigation and support for emotionally vulnerable users who may be using the chatbot as a platform for expressing their distressing emotions.

Robustness Check: We conducted ANOVA tests across different user groups (grouping users every 10th percentile), including superusers, to assess the impact of user ranking on linguistic

Table 2. The Welch's t-test results examine the quantified differences in user characteristics and evaluation of chatbot capabilities between active and general users from the survey.

Variables (Welch’s t-test)	t	p	d	Super (n=39)		General (n=37)	
				Mean	(SD)	Mean	(SD)
Mental Dispositions							
Empathy	2.31	<.05*	0.53	3.86	(0.55)	3.54	(0.67)
Anthropomorphism	5.48	<.001***	1.26	3.80	(0.84)	2.76	(0.81)
Depression	3.20	<.01**	0.73	12.67	(7.12)	7.65	(6.50)
Perceptions of Chatbot							
Likability	0.23	0.82	0.05	3.35	(0.83)	3.30	(1.08)
Intelligence	1.23	0.22	0.28	3.45	(0.92)	3.18	(0.94)
Truthfulness	0.49	0.63	0.11	2.97	(1.29)	2.84	(1.17)

Note 1: t: t-statistics, p: p-value, d: Cohen's d-value, SD: standard deviation

Note 2: Refer to the detailed stats by the demographic variables' category in the Appendix

styles and topic preferences previously discussed. The ANOVA findings confirm our prior analyses, revealing significant differences ($p < .001$) between superusers and other user groups in specific LIWC categories identified in the t-tests as having substantial effect size (refer to Figures 7 and 8). The post hoc Tukey test indicates that, except for superusers, most user groups exhibit overlapping LIWC scores (non-significant difference with adjusted $p > .05$). Additionally, an exclusive parallel is observed between the 90-99th percentile users and the 99-100th percentile superusers, particularly in categories such as *Positive*, *Sadness*, and *Sexual*, supporting our decision to set the threshold for general users before the 90th percentile. Similarly, the ANOVA on BERTopic results mirrors consistent trends, with topics found to be significantly different between two groups in t-tests also differing significantly across user groups in the ANOVA analysis, as detailed in the Appendix (see Figure 9). These ANOVA results underscore the study's consistency.

4.2 User Dispositions of Superusers

We employed Welch's t-test to determine disparities in mental dispositions of superusers and general users. We opted for Welch's t-test to accommodate any deviations in our data from normal distributions. In addition, we conducted a Fisher's exact test to assess demographic differences between the two groups to categorical variables, such as gender, education level, and race. The Table 2 summarizes the main findings.

Superusers displayed a higher level of empathy compared to general users ($p < .05$, super: $M=3.86$, $SD=0.55$, general: $M=3.54$, $SD=0.67$), and they also had a greater propensity to anthropomorphize non-human agents ($p < .001$, super: $M=3.80$, $SD=0.84$, general: $M=2.76$, $SD=0.81$). This difference in anthropomorphism yielded a substantial magnitude, with a Cohen's value of 1.26, which is considered to be a large effect size ($d > 0.8$).

In the survey results of the Patient Health Questionnaire-9 (PHQ-9) tool, we observed a notable difference in the average depression levels between the two groups ($p < .01$ with Cohen's $d = 0.73$). The general user group had a mean depression score of 7.65 ($SD=6.50$), indicating a state of normal to mild depression. Conversely, the mean depression score for superusers was 12.67 ($SD=7.12$), signifying a moderate level of depression according to PHQ-9's interpretation criteria. As noted earlier, we consider this score to represent the general mood of users at the time of the survey rather than their clinical states.

In user perception of chatbot capabilities, there were no statistically significant differences in mean values between superusers and general users when it came to rating the likability ($p = 0.64$), intelligence ($p = 0.12$), or truthfulness ($p = 0.56$) of the chatbot. The mean age of all participants was 23.9 ($SD=8.99$). Superusers had a mean age of 23.77 ($SD=9.30$), and the general user group had

Table 3. Summary of findings on user behavior and perception statements, as well as the prevailing group for each statement between the most active users and general users.

User Behavior	Prevailing Group	User Perception	Prevailing Group
Friendly tone to chatbot:	Top Active Users	Chatbot is human-like:	Top Active Users
Use of offensive language:	General Users	Chatbot is likable:	Similar
Talk about family issues:	Top Active Users	Chatbot is trustworthy:	Similar
Higher depression scores:	Top Active Users	Chatbot is intelligent:	Similar

a mean age of 24.05 (SD=8.77), with no significant age difference ($p = 0.89$). Educational attainment did not significantly differ between the two groups ($p = 0.30$), with most users having at least a high school degree. There were no significant variations in gender ($p = 0.76$) and race ($p = 0.20$) distributions between the two groups. Figure 4 and 5 in the Appendix provide detailed distribution information within each category of age, gender, education, and race.

5 Discussion

5.1 Anthropomorphizing Chatbots

Our research findings suggest that individuals who are highly engaged with the SimSimi chatbot tend to empathize more with non-human agents and are more likely to humanize them. Table 3 summarizes the survey results. Top active users adopt a friendly and less toxic conversational style, as evidenced by their significantly reduced use of profanity, anger, and sex-related words during chats. They also exhibit a chat pattern that involves a higher use of third-person pronouns when mentioning SimSimi, potentially to encourage interactive conversations [8]. These inclinations were consistent with our survey, where top active users exhibited a stronger tendency of empathy compared to general users. This friendlier behavior deviates from the trend observed in previous studies, such as the one conducted by Hill et al. [47], which found a higher prevalence of profanity and toxicity in user interactions. However, this is consistent with the findings of Darling et al. [30], whose research showed that individuals characterized by higher levels of empathy exhibited a greater reluctance to engage in aggressive behavior toward the robots. Thus, the most active users may have developed a sense of companionship with the chatbot.

Despite high empathy, top active users were less interested in the chatbot and its technical capabilities. The linguistic analysis did not indicate an increased utilization of second-person pronouns directed towards SimSimi and topic analysis revealed fewer discussions about the chat functionalities. This suggests that users were more indifferent to the system’s technical aspects and showed minimal inclination to challenge its nature by omitting questions like “Are you real?” and “Are you a robot?” While treating SimSimi politely as a confidant, top active users’ survey responses rated its intelligence and truthfulness similarly to general users.

This pattern can be explained by top active users’ strong anthropomorphism, as revealed through the survey. Users with limited knowledge of nonhuman entities, and decreased motivation to explain and understand their behavior tend to anthropomorphize artificial agents [35]. Based on a prior study demonstrating heightened cooperation with agents that did not reveal their artificial nature [50], it seems likely that the mental model of anthropomorphizing a conversational agent is integral to fostering engaging and enduring interactions with a bot. However, an increasing body of literature points to the potential negative effects of anthropomorphism. Previous research has indicated that the extent to which individuals anthropomorphize an agent significantly impacts interactions between humans and agents, often leading to less favorable outcomes for the user. For example, Shin and Kim [82] studied the relationship between individual loneliness and anthropomorphism. They observed that individuals who attributed more human-like traits to nonhuman entities experienced

higher levels of self-rated loneliness. Other studies [13, 101] have also shown that lonely people wanted to keep social connections and made up for their lack of social connections by giving human-like qualities to non-human things.

Previous studies have found a link between an increasing tendency to anthropomorphize non-human entities and a rise in dehumanization in society [82, 100, 101]. For example, people may feel attached to human-like technology and reduce face-to-face communication as they establish an attachment to artificial entities [100]. This trend may make individuals see other humans as distant [43]. According to prior research [100], dehumanization leads to diminished moral involvement, consequently heightening aggression towards both individuals and groups [11, 90], fostering support for discrimination against racial outgroups [39], and promoting negative overall attitudes towards outgroups [48].

5.2 Intimacy Leading to Vulnerability

As shown in Table 3, top active users (superuser), with their empathetic tendencies and strong affinity for anthropomorphism, appear to be naturally inclined toward self-disclosure when interacting with social chatbots. Previous research on social media platforms has indicated that prolonged engagement can lead to privacy concerns through heightened self-disclosure [72]. The analysis of linguistic patterns and conversation topics consistently demonstrates a trend of heightened self-disclosure among the most active SimSimi users. They are more open to discussing personal matters like their health, family, and homes compared to general users, with their word usage surpassing that of the average Twitter user in LIWC's social categories ($M=0.52$, $SD=0.87$ for *Family* and $M=0.26$, $SD=0.43$ for *Home*[16]), underscoring their remarkable willingness to engage in such discussions ($p < .001$). They also exchanged their views on social discrimination, a topic known to elicit strong emotional responses and a heightened need for both support and information [24, 70]. Experimental investigations into the significance of LIWC scores have also demonstrated that a greater proportion of social and personal concern categories is associated with more truthful behavior [44].

Top active users also tend to use language associated with depressive moods and openly share their feelings of sadness with the chatbot. According to the LIWC analysis, they used more negative affective words, such as anxiety and sadness, than general users. The BERTopic analysis revealed that the most active users' messages exhibited a higher prevalence of the 'Sadness' topic, which encompasses references to self-harm or suicidal thoughts, as well as the user expressing negative emotions. This matches the results of our user survey, which showed that the most active SimSimi users' self-reported depressive mood scores were at a moderate level, higher than those of the general users. Moreover, the analysis of time orientation suggests that SimSimi superusers predominantly focus on the present at the expense of the future, which was behaviorally linked to impulsivity and less hesitancy to take risks [52, 55, 110]. These findings collectively raise important questions about the emotional well-being of top active chatbot users and the potential risks associated with their prolonged interactions with the chatbot.

We are unable to draw definitive conclusions about whether extended chatbot interactions caused these characteristics in top active chatbot users or if these traits were inherent from the outset. The link between chat frequency, empathy, anthropomorphism, and depressive moods could have existed before the chatbot was adopted or emerged over the course of three years of use. Similar debates are found in studies on problematic social network use, where some researchers blame negative mental traits on users' intrinsic characteristics [64], while others stress the role of social cues in triggering unhealthy online behavior [4]. Some research suggests that users' inherent traits may shape their interactions with autonomous agents [30]. Future chatbot users who are familiar with more advanced language models, such as ChatGPT or Gemini, may interact with machines

differently from the patterns we observe in this work. Nonetheless, understanding the needs of highly active users and potential risks to vulnerable groups remains crucial for ethical chatbot system adoption.

6 Design Implications

6.1 How to Protect Vulnerable Users

Contrary to the prevailing thinking that chatbot conversations are more synthetic and toxic compared to human-human interactions [69], our data revealed a different trend. We observed that top active users formed “social” interactions with the chatbot. These individuals, who exhibited high levels of empathy and anthropomorphism towards machines, engaged in intimate dialogues with the chatbot and often shared personal and self-disclosing information. These users also reported higher levels of depressive mood than general users and were more likely to express negative emotions to the chatbot.

Our findings align with research in the medical domain, where users felt more comfortable disclosing mental conditions to virtual humans during clinical interviews, viewing them as “safe” interaction partners [40]. Despite SimSimi’s lack of a specific medical design, we observed that highly active users developed empathetic feelings towards the chatbot and experienced emotional relief. However, it is essential to balance the potential benefits and risks. Strong attachments to nonhuman entities may weaken the motivation for human interactions [82], potentially leading to dehumanization and moral disengagement, which can enable unethical behavior toward others [100]. Children and older adults tend to anthropomorphize non-human agents compared to younger adults [35, 67], and the risks associated with excessive dependence on chatbots may have more impact on these vulnerable populations.

Given this risk, developers should provide users with clear information about the limitations of chatbots to prevent over-reliance. When users have accessible knowledge, they are more likely to form a realistic mental model of non-human entities and less likely to anthropomorphize machines [35]. Therefore, chatbot developers can encourage users to explore the chatbot’s capabilities to form a more accurate understanding of these systems. This approach could involve integrating engaging features such as games and quizzes into the system promoting informative interactions between users and chatbots. Such an approach may prove particularly beneficial for younger and older user demographics [6, 81].

6.2 How to Offer Privacy Guidelines

Prolonged engagement with social chatbots increases the likelihood of users sharing more information with the system. Similar to extended use of social media [72], our top active users may become less cautious in safeguarding their personal information, potentially due to their emotional vulnerability and the increased need for self-disclosure. Therefore, the responsibility for protecting leaked information should not solely rest with the users, as they might not fully grasp the consequences of their actions. It falls upon system developers and data custodians to take essential steps to ensure their users’ privacy. Guidelines should extend beyond technical measures like data filtering to prevent inadvertently incorporating sensitive information into training. Chatbot interfaces should be designed to discourage users from engaging in risky behavior and promote responsible information sharing.

Chatbots should proactively inform users about the possibility of their conversations being accessible to both developers and third parties. Currently, many conversational AI systems, including widely used models like ChatGPT, do not provide explicit disclosures regarding using user chat histories [107]. Chatbot platforms may not always guarantee the security of their confidential information. Knowing that their data could be exposed makes users more cautious about managing

their sensitive information. To achieve transparency, chatbot platforms can implement features such as highlighting potentially sensitive data for users or prompting the chatbot to remind users about the importance of safeguarding their personal information. An alternative solution is to enable users to control their personal information. For instance, a chatbot service could provide a "review board" that displays the personal information and sensitive topics the user disclosed during the chats (e.g., user names, age, hobbies, residing city, and sexual preference). Users could then decide whether to keep such information in the chatbot for personalized service or to delete it.

6.3 Limitations and Future research

Future work can complement this study by examining the evolution of users' emotional experiences and interaction dynamics over time. Our analyses reveal statistical differences in perceived mental dispositions toward the chatbot among different user groups. However, quantitatively determining whether chatbot usage frequency influences user perceptions or vice versa remains a challenging aspect. Conducting additional experiments and interviews may provide a deeper understanding of the causal relationships within the observed patterns, which the current work lacks. Longitudinal controlled studies can shed light on changes in users' emotional states and how their interactions with chatbots contribute to these changes.

User emotional states, as captured by the LIWC tool, may influence their choice of conversation topics with the chatbot, and conversely, the topics discussed may also impact users' emotional states. While not within our study's scope, this area is crucial for deeper insights into user interaction preferences. Future research could employ alternative statistical methods to explore these interactions among the variables we studied.

Our study employed text analysis and surveys to analyze users, yet differences in timing between these methods may impact results. Furthermore, we acknowledge that users who completed the survey may possess different characteristics than those who did not, potentially impacting the significance of our findings. General users who completed the survey may have a higher likelihood of transitioning to superuser status in the future, given their heightened attention to the app. Future studies could mitigate this by integrating survey questions during interactions.

Despite Simsimi's relatively simple design, data indicate that some users have developed long-term relationships with the chatbot. As chatbots evolve and progress towards human-like conversational abilities, we anticipate that the behaviors we observed will become more prominent and complex. Moreover, given the diversity of designs and functionalities among chatbots with large language models, it is crucial to investigate potential cultural and platform variations that may influence user engagement. The observed user behavior patterns may differ across various countries and languages when engaging with a chatbot. Future research on the Simsimi platform, which spans a large global user base, could directly compare its results with our findings. This comparative analysis would shed light on the generalizability of our conclusions and the influence of cultural and linguistic factors on chatbot interactions. Addressing these dimensions can better support users, thereby laying the foundation for more robust guidelines in the development of reliable chatbot systems.

7 Conclusion

This study aimed to characterize the behaviors and perceptions of chatbot superusers who are highly engaged in using chatbots. We measured unique behaviors or perceptions of highly active users towards the chatbot through linguistic analysis and topic modeling and evaluated their dispositional profiles through a voluntary survey. We discovered superusers might be vulnerable individuals who express higher levels of empathy, anthropomorphism, and depressive moods. They

developed polite and friendly relationships with the chatbot and were more willing to self-disclose their personal lives and emotional struggles compared to other users.

These patterns underscore the potential role of social chatbots in providing professional emotional support. However, challenges may arise, including a diminished willingness to interact with other humans (dehumanization) and the risk of private information leakage when developing highly engaging and human-like companions. As the landscape of chatbot interactions continues to evolve, recognizing the emotional needs of superusers becomes crucial for creating a supportive environment that prioritizes users' privacy and well-being. While our study focused on a single chatbot, investigating other platforms and adapting bot behavior remain pertinent areas for future exploration.

8 Acknowledgement

This research was supported by the Institute for Basic Science (IBS-R029-C2) and the National Research Foundation of Korea grant (RS-2022-00165347). This research was done while H. Chin and M. Cha were affiliated with IBS. M. Cha is also affiliated with KAIST. The authors greatly thank SimSimi Inc. and CEO Mr. J. Choi for providing the data for research purposes. We also thank the anonymous reviewers and the chairs for their insightful reviews.

References

- [1] 2019. Much More Than a Chatbot: China's Xiaoice Mixes AI With Emotions and Wins Over Millions of Fans - Microsoft Stories Asia. Retrieved May 30, 2023 from <http://tinyurl.com/2p95tzhw>
- [2] 2022. LIWC-22. Retrieved May 30, 2023 from <https://www.liwc.app/>
- [3] 2023. *Perspective API*. Retrieved March 30, 2023 from <https://perspectiveapi.com/>
- [4] Irum Abbasi and Michelle Drouin. 2019. Neuroticism and Facebook Addiction: How Social Media Can Affect Mood? *The American Journal of Family Therapy* 47, 4 (2019), 199–215.
- [5] Abhishek Aggarwal, Cheuk Chi Tam, Dezhi Wu, Xiaoming Li, and Shan Qiao. 2023. Artificial Intelligence-Based Chatbots for Promoting Health Behavioral Changes: Systematic Review. *Journal of Medical Internet Research* 25 (2023), e40789.
- [6] Maximilian Altmeyer, Pascal Lessel, and Antonio Krüger. 2018. Investigating Gamification for Seniors Aged 75+. In *Proceedings of the Designing Interactive Systems Conference*.
- [7] Theo Araujo. 2018. Living up to the Chatbot Hype: The Influence of Anthropomorphic Design Cues and Communicative Agency Framing on Conversational Agent and Company Perceptions. *Computers in Human Behavior* 85 (2018), 183–189.
- [8] Jaime Arguello, Brian S Butler, Elisabeth Joyce, Robert Kraut, Kimberly S Ling, Carolyn Rosé, and Xiaoqing Wang. 2006. Talk to Me: Foundations for Successful Individual-Group Interactions in Online Communities. In *Proceedings of the SIGCHI conference on Human Factors in computing systems*.
- [9] Imane El Atillah. 2023. *AI Chatbot Blamed for "Encouraging" Young Father to Take His Own Life*. Retrieved May 30, 2023 from <https://bit.ly/3Nj7Jlk>
- [10] Petter Bae Bae Brandtzæg, Marita Skjuve, Kim Kristoffer Kristoffer Dysthe, and Asbjørn Følstad. 2021. When the Social Becomes Non-Human: Young People's Perception of Social Support in Chatbots. In *Proceedings of the ACM SIGCHI Conference on Human Factors in Computing Systems*.
- [11] Albert Bandura, Bill Underwood, and Michael E Fromson. 1975. Disinhibition of Aggression Through Diffusion of Responsibility and Dehumanization of Victims. *Journal of Research in Personality* 9, 4 (1975), 253–269.
- [12] Christoph Bartneck, Dana Kulić, Elizabeth Croft, and Susana Zoghbi. 2009. Measurement Instruments for the Anthropomorphism, Animacy, Likeability, Perceived Intelligence, and Perceived Safety of Robots. *International Journal of Social Robotics* 1 (2009), 71–81.
- [13] Jennifer A Bartz, Kristina Tchalova, and Can Fenerci. 2016. Reminders of Social Connection Can Attenuate Anthropomorphism: A Replication and Extension of Epley, Akalis, Waytz, and Cacioppo (2008). *Psychological Science* 27, 12 (2016), 1644–1650.
- [14] Clare Beatty, Tanya Malik, Saha Meheli, and Chaitali Sinha. 2022. Evaluating the Therapeutic Alliance With a Free-Text Cbt Conversational Agent (Wysa): A Mixed-Methods Study. *Frontiers in Digital Health* 4 (2022), 847991.

- [15] Ivo Benke, Michael Thomas Knierim, and Alexander Maedche. 2020. Chatbot-Based Emotion Management for Distributed Teams: A Participatory Design Study. *Proceedings of the ACM on Human-Computer Interaction* 4, CSCW2 (2020), 1–30.
- [16] Ryan L Boyd, Ashwini Ashokkumar, Sarah Seraj, and James W Pennebaker. 2022. The Development and Psychometric Properties of Liwc-22. *Austin, TX: University of Texas at Austin* (2022), 1–47.
- [17] Petter Bae Brandtzaeg and Asbjørn Følstad. 2017. Why People Use Chatbots. *Internet Science* (2017), 377–392.
- [18] Petter Bae Brandtzaeg and Asbjørn Følstad. 2018. Chatbots: Changing User Needs and Motivations. *Interactions* 25, 5 (2018), 38–43.
- [19] Bonnie Canziani and Sara MacSween. 2021. Consumer Acceptance of Voice-Activated Smart Home Devices for Product Information Seeking and Online Ordering. *Computers in Human Behavior* 119 (2021), 106714.
- [20] Richard Catrambone, John Stasko, and Jun Xiao. 2004. *ECA as User Interface Paradigm*. Springer Netherlands, Dordrecht, 239–267.
- [21] Benjamin Chaix, Jean-Emmanuel Bibault, Arthur Pienkowski, Guillaume Delamon, Arthur Guillemassé, Pierre Nectoux, Benoît Brouard, et al. 2019. When Chatbots Meet Patients: One-Year Prospective Study of Conversations Between Patients With Breast Cancer and a Chatbot. *JMIR cancer* 5, 1 (2019), e12856.
- [22] Yang Cheng and Hua Jiang. 2020. How Do AI-Driven Chatbots Impact User Experience? Examining Gratifications, Perceived Privacy Risk, Satisfaction, Loyalty, and Continued Use. *Journal of Broadcasting & Electronic Media* 64, 4 (2020), 592–614.
- [23] Hyojin Chin, Gabriel Lima, Mingi Shin, Assem Zhunis, Chiyoung Cha, Junghoi Choi, and Meeyoung Cha. 2023. User-Chatbot Conversations During the COVID-19 Pandemic: Study Based on Topic Modeling and Sentiment Analysis. *Journal of Medical Internet Research* 25 (2023), e40922.
- [24] Hyojin Chin, Gabriel Lima, Mingi Shin, Assem Zhunis, Chiyoung Cha, Junghoi Choi, and Meeyoung Cha. 2023. User-Chatbot Conversations During the COVID-19 Pandemic: Study Based on Topic Modeling and Sentiment Analysis. *J Med Internet Res* 25 (27 Jan 2023), e40922.
- [25] Hyojin Chin, Lebogang Wame Molefi, and Mun Yong Yi. 2020. Empathy Is All You Need: How a Conversational Agent Should Respond to Verbal Abuse. In *Proceedings of the ACM SIGCHI Conference on Human Factors in Computing Systems*.
- [26] Hyojin Chin, Hyeonho Song, Gumhee Baek, Mingi Shin, Chani Jung, Meeyoung Cha, Junghoi Choi, and Chiyoung Cha. 2023. The Potential of Chatbots for Emotional Support and Promoting Mental Well-Being in Different Cultures: Mixed Methods Study. *Journal of Medical Internet Research* 25 (2023), e51712.
- [27] Noel K Clark and Derek R Rutter. 1985. Social Categorization, Visual Cues, and Social Judgements. *European Journal of Social Psychology* 15, 1 (1985), 105–119.
- [28] Emmelyn AJ Croes and Marjolijn L Antheunis. 2021. Can We Be Friends With Mitsuku? A Longitudinal Study on the Process of Relationship Formation Between Humans and a Social Chatbot. *Journal of Social and Personal Relationships* 38, 1 (2021), 279–300.
- [29] Florian Daniel, Maristella Matera, Vittorio Zaccaria, and Alessandro Dell’Orto. 2018. Toward Truly Personal Chatbots: On the Development of Custom Conversational Assistants. In *Proceedings of the IEEE International Conference on Healthcare Informatics Workshop*.
- [30] Kate Darling, Palash Nandy, and Cynthia Breazeal. 2015. Empathic Concern and the Effect of Stories in Human-Robot Interaction. In *Proceedings of the IEEE International Symposium on Robot and Human Interactive Communication*.
- [31] Mauro De Gennaro, Eva G Krumhuber, and Gale Lucas. 2020. Effectiveness of an Empathic Chatbot in Combating Adverse Effects of Social Exclusion on Mood. *Frontiers in psychology* 10 (2020), 3061.
- [32] Ewart J De Visser, Samuel S Monfort, Ryan McKendrick, Melissa AB Smith, Patrick E McKnight, Frank Krueger, and Raja Parasuraman. 2016. Almost Human: Anthropomorphism Increases Trust Resilience in Cognitive Agents. *Journal of Experimental Psychology: Applied* 22, 3 (2016), 331.
- [33] Gerrit De Vynck. 2023. ChatGPT Loses Users for First Time, Shaking Faith in AI Revolution. (7 2023). Retrieved May 30, 2023 from <https://www.washingtonpost.com/technology/2023/07/07/chatgpt-users-decline-future-ai-openai/>
- [34] Gilly Dosovitsky and Eduardo Bunge. 2023. Development of a Chatbot for Depression: Adolescent Perceptions and Recommendations. *Child and Adolescent Mental Health* 28, 1 (2023), 124–127.
- [35] Nicholas Epley, Adam Waytz, and John T Cacioppo. 2007. On Seeing Human: A Three-Factor Theory of Anthropomorphism. *Psychological Review* 114, 4 (2007), 864.
- [36] Kathleen Kara Fitzpatrick, Alison Darcy, and Molly Vierhile. 2017. Delivering Cognitive Behavior Therapy to Young Adults With Symptoms of Depression and Anxiety Using a Fully Automated Conversational Agent (Woebot): A Randomized Controlled Trial. *JMIR Mental Health* 4, 2 (2017), e7785.
- [37] Silvia García-Méndez, Francisco De Arriba-Pérez, Francisco J González-Castaño, JOSé A Regueiro-Janeiro, and Felipe Gil-Castiñeira. 2021. Entertainment Chatbot for the Digital Inclusion of Elderly People Without Abstraction Capabilities. *IEEE Access* 9 (2021), 75878–75891.

- [38] Ulrich Gnewuch, Stefan Morana, and Alexander Maedche. 2017. Towards Designing Cooperative and Social Conversational Agents for Customer Service. In *Proceedings of the International Conference on Information Systems*.
- [39] Phillip Atiba Goff, Jennifer L Eberhardt, Melissa J Williams, and Matthew Christian Jackson. 2008. Not Yet Human: Implicit Knowledge, Historical Dehumanization, and Contemporary Consequences. *Journal of Personality and Social Psychology* 94, 2 (2008), 292.
- [40] Jonathan Gratch, Gale M Lucas, Aisha Aisha King, and Louis-Philippe Morency. 2014. It's Only a Computer: The Impact of Human-agent Interaction in Clinical Interviews. In *Proceedings of the International Conference on Autonomous Agents and Multiagent Systems*.
- [41] Erin Griffith and Cade Metz. 2023. A New Area of AI Booms, Even Amid the Tech Gloom.
- [42] Maarten Grootendorst. 2020. BERTopic: Leveraging BERT and c-Tf-Idf to Create Easily Interpretable Topics.
- [43] Tal Orian Harel, Jessica Katz Jameson, and Ifat Maoz. 2020. The Normalization of Hatred: Identity, Affective Polarization, and Dehumanization on Facebook in the Context of Intractable Political Conflict. *Social Media+ Society* 6, 2 (2020), 2056305120913983.
- [44] Md Kamrul Hasan, Taylan Sen, Yiming Yang, Raiyan Abdul Baten, Kurtis Glenn Haut, and Mohammed Ehsan Hoque. 2019. LIWC into the Eyes: Using Facial Features to Contextualize Linguistic Analysis in Multimodal Communication. In *Proceedings of the IEEE International Conference on Affective Computing and Intelligent Interaction*.
- [45] Béatrice S Hasler, Peleg Tuchman, and Doron Friedman. 2013. Virtual Research Assistants: Replacing Human Interviewers by Automated Avatars in Virtual Worlds. *Computers in Human Behavior* 29, 4 (2013), 1608–1616.
- [46] Yuhao He, Li Yang, Xiaokun Zhu, Bin Wu, Shuo Zhang, Chunlian Qian, and Tian Tian. 2022. Mental Health Chatbot for Young Adults With Depressive Symptoms During the COVID-19 Pandemic: Single-Blind, Three-Arm Randomized Controlled Trial. *Journal of Medical Internet Research* 24, 11 (2022), e40719.
- [47] Jennifer Hill, W Randolph Ford, and Ingrid G Farreras. 2015. Real Conversations With Artificial Intelligence: A Comparison Between Human–Human Online Conversations and Human–Chatbot Conversations. *Computers in Human Behavior* 49 (2015), 245–250.
- [48] Gordon Hodson and Kimberly Costello. 2007. Interpersonal Disgust, Ideological Orientations, and Dehumanization as Predictors of Intergroup Attitudes. *Psychological Science* 18, 8 (2007), 691–698.
- [49] Carolin Ischen, Janice Butler, and Jakob Ohme. 2023. Chatting About the Unaccepted: Self-Disclosure of Unaccepted News Exposure Behaviour to a Chatbot. *Behaviour & Information Technology* 0, 0 (2023), 1–13.
- [50] Fatimah Ishowo-Oloko, Jean-François Bonnefon, Zakariyah Soroye, Jacob Crandall, Iyad Rahwan, and Talal Rahwan. 2019. Behavioural Evidence for a Transparency–Efficiency Tradeoff in Human–Machine Cooperation. *Nature Machine Intelligence* 1, 11 (2019), 517–521.
- [51] Eunkyoung Jo, Daniel A Epstein, Hyunhoon Jung, and Young-Ho Kim. 2023. Understanding the Benefits and Challenges of Deploying Conversational AI Leveraging Large Language Models for Public Health Intervention. In *Proceedings of the ACM SIGCHI Conference on Human Factors in Computing Systems*.
- [52] Łukasz Jochemczyk, Janina Pietrzak, Rafał Buczkowski, Maciej Stolarski, and Łukasz Markiewicz. 2017. You Only Live Once: Present-Hedonistic Time Perspective Predicts Risk Propensity. *Personality and Individual Differences* 115 (2017), 148–153.
- [53] Jodie. 2023. What ChatGPT's New Features Mean For Your Business. Retrieved Jan 09, 2024 from <https://www.forbes.com/sites/jodiecook/2023/11/07/what-chatgpts-new-features-mean-for-your-business/?sh=6b8893202b87>
- [54] Samantha Murphy Kelly. 2023. Snapchat's New AI Chatbot Is Already Raising Alarms Among Teens and Parents. Retrieved May 30, 2023 from <https://edition.cnn.com/2023/04/27/tech/snapchat-my-ai-concerns-wellness/index.html>
- [55] Kelli A Keough, Philip G Zimbardo, and John N Boyd. 1999. Who's Smoking, Drinking, and Using Drugs? Time Perspective as a Predictor of Substance Use. *Basic and applied social psychology* 21, 2 (1999), 149–164.
- [56] Kurt Kroenke, Robert L Spitzer, and Janet BW Williams. 2001. The PHQ-9: Validity of a Brief Depression Severity Measure. *Journal of General Internal Medicine* 16, 9 (2001), 606–613.
- [57] Mirosława Lasek and Szymon Jessa. 2013. Chatbots for Customer Service on Hotels' Websites. *Information Systems in Management* 2, 2 (2013), 146–158.
- [58] Yi-Chieh Lee, Naomi Yamashita, Yun Huang, and Wai Fu. 2020. "I Hear You, I Feel You": Encouraging Deep Self-Disclosure through a Chatbot. In *Proceedings of the ACM SIGCHI Conference on Human Factors in Computing Systems*.
- [59] Jindong Leo-Liu. 2023. Loving a "Defiant" AI Companion? The Gender Performance and Ethics of Social Exchange Robots in Simulated Intimate Interactions. *Computers in Human Behavior* 141 (2023), 107620.
- [60] Brooke Levis, Andrea Benedetti, and Brett D Thombs. 2019. Accuracy of Patient Health Questionnaire-9 (Phq-9) for Screening to Detect Major Depression: Individual Participant Data Meta-Analysis. *BMJ* 365 (2019).
- [61] Jindong Liu. 2021. Social Robots as the Bride? Understanding the Construction of Gender in a Japanese Social Robot Product. *Human-Machine Communication* 2 (2021), 105–120.
- [62] Gustavo López, Luis Quesada, and Luis A. Guerrero. 2018. Alexa vs. Siri vs. Cortana vs. Google Assistant: A Comparison of Speech-Based Natural User Interfaces. In *Advances in Human Factors and Systems Interaction*, Isabel L. Nunes (Ed.).

- [63] Gale M Lucas, Jonathan Gratch, Aisha King, and Louis-Philippe Morency. 2014. It's Only a Computer: Virtual Humans Increase Willingness to Disclose. *Computers in Human Behavior* 37 (2014), 94–100.
- [64] Davide Marengo, Ilaria Poletti, and Michele Settanni. 2020. The Interplay Between Neuroticism, Extraversion, and Social Media Addiction in Young Adult Facebook Users: Testing the Mediating Role of Online Activity Using Objective Data. *Addictive behaviors* 102 (2020), 106150.
- [65] Lenin Medeiros, Tibor Bosse, and Charlotte Gerritsen. 2021. Can a Chatbot Comfort Humans? Studying the Impact of a Supportive Chatbot on Users' Self-Perceived Stress. *Transactions on Human-Machine Systems* 52, 3 (2021), 343–353.
- [66] Tripp Mickle, Cade Metz, and Nico Grant. 2023. The Chatbots Are Here, and the Internet Industry Is in a Tizzy. Retrieved May 30, 2023 from <https://tinyurl.com/k6hc846e>
- [67] Lucas Morillo-Mendez, Martien GS Schrooten, Amy Loutfi, and Oscar Martinez Mozos. 2022. Age-Related Differences in the Perception of Robotic Referential Gaze in Human-Robot Interaction. *International Journal of Social Robotics* (2022), 1–13.
- [68] Sara Morrison. 2023. Is the AI boom already over? Retrieved May 30, 2023 from <https://tinyurl.com/5n7xaw5r>
- [69] Yi Mou and Kun Xu. 2017. The Media Inequality: Comparing the Initial Human-Human and Human-AI Social Interactions. *Computers in Human Behavior* 72 (2017), 432–440.
- [70] Marcia Mundt, Karen Ross, and Charla M Burnett. 2018. Scaling Social Movements Through Social Media: The Case of Black Lives Matter. *Social Media+ Society* 4, 4 (2018), 2056305118807911.
- [71] Vojtěch Mýlek, Lenka Dedkova, and Alexander P Schouten. 2023. Adolescents' Online Communication and Self-Disclosure to Online and Offline Acquaintances: Differential Effects of Social Anxiety and Depressed Moods. *Journal of Media Psychology: Theories, Methods, and Applications* (2023).
- [72] Sina Ostendorf, Silke M Müller, and Matthias Brand. 2020. Neglecting Long-Term Risks: Self-Disclosure on Social Media and Its Relation to Individual Decision-Making Tendencies and Problematic Social-Networks-Use. *Frontiers in Psychology* 11 (2020), 2913.
- [73] Shweta Paliwal, Vishal Bharti, and Amit Kumar Mishra. 2020. AI Chatbots: Transforming the Digital World. *Recent Trends and Advances in Artificial Intelligence and Internet of Things* (2020), 455–482.
- [74] Katherine Panciera, Aaron Halfaker, and Loren Terveen. 2009. Wikipedians Are Born, Not Made: A Study of Power Editors on Wikipedia. In *Proceedings of the ACM International Conference on Supporting Group Work*.
- [75] James W Pennebaker, Matthias R Mehl, and Kate G Niederhoffer. 2003. Psychological Aspects of Natural Language Use: Our Words, Our Selves. *Annual Review of Psychology* 54, 1 (2003), 547–577.
- [76] Matthew D Pickard, Catherine A Roster, and Yixing Chen. 2016. Revealing Sensitive Information in Personal Interviews: Is Self-Disclosure Easier With Humans or Avatars and Under What Conditions? *Computers in Human Behavior* 65 (2016), 23–30.
- [77] Hans-Peter Piepho. 2004. An Algorithm for a Letter-Based Representation of All-Pairwise Comparisons. *Journal of Computational and Graphical Statistics* (2004), 456–466.
- [78] Giovanni Quattrone, Licia Capra, and Pasquale De Meo. 2015. There's No Such Thing as the Perfect Map: Quantifying Bias in Spatial Crowd-Sourcing Datasets. In *Proceedings of the ACM Conference on Computer Supported Cooperative Work & Social Computing*.
- [79] Andrew R Chow. 2023. AI-Human Romances Are Flourishing—And This Is Just the Beginning. Retrieved Jan 09, 2024 from <https://time.com/6257790/ai-chatbots-love/>
- [80] Arsénio Reis, Dennis Paulino, Hugo Paredes, Isabel Barroso, Maria João Monteiro, Vitor Rodrigues, and João Barroso. 2018. Using Intelligent Personal Assistants to Assist the Elderlies an Evaluation of Amazon Alexa, Google Assistant, Microsoft Cortana, and Apple Siri. In *Proceedings of the IEEE International Conference on Technology and Innovation in Sports, Health and Wellbeing*.
- [81] Katie Seaborn and Deborah I Fels. 2015. Gamification in Theory and Action: A Survey. *International Journal of human-computer studies* 74 (2015), 14–31.
- [82] Hong Im Shin and Juyoung Kim. 2020. My Computer Is More Thoughtful Than You: Loneliness, Anthropomorphism and Dehumanization. *Current Psychology* 39 (2020), 445–453.
- [83] Redwood Shores. 2020. Global Study: 82% of People Believe Robots Can Support Their Mental Health Better Than Humans. Retrieved Jan 12, 2024 from <https://www.oracle.com/news/announcement/ai-at-work-100720/>
- [84] Nakatani Shuyo. 2021. *langdetect 1.0.9*. Retrieved May 30, 2023 from <https://pypi.org/project/langdetect/>
- [85] Ben Singh, Timothy Olds, Jacinta Brinsley, Dot Dumuid, Rosa Virgara, Lisa Matricciani, Amanda Watson, Kimberley Szeto, Emily Eglitis, Aaron Miatke, et al. 2023. Systematic Review and Meta-Analysis of the Effectiveness of Chatbots on Lifestyle Behaviours. *npj Digital Medicine* 6, 1 (2023), 118.
- [86] Marita Skjuve, Asbjørn Følstad, Knut Inge Fostervold, and Petter Bae Brandtzaeg. 2021. My Chatbot Companion—a Study of Human-Chatbot Relationships. *International Journal of Human-Computer Studies* 149 (2021), 102601.
- [87] Marita Skjuve, Asbjørn Følstad, Knut Inge Fostervold, and Petter Bae Brandtzaeg. 2022. A Longitudinal Study of Human-Chatbot Relationships. *International Journal of Human-Computer Studies* 168 (2022), 102903.

- [88] R Nathan Spreng*, Margaret C McKinnon*, Raymond A Mar, and Brian Levine. 2009. The Toronto Empathy Questionnaire: Scale development and initial validation of a factor-analytic solution to multiple empathy measures. *Journal of Personality Assessment* 91, 1 (2009), 62–71.
- [89] Georges Steffgen, Andreas König, Jan Pfetsch, and André Melzer. 2011. Are Cyberbullies Less Empathic? Adolescents' Cyberbullying Behavior and Empathic Responsiveness. *Cyberpsychology, Behavior, and Social Networking* 14, 11 (2011), 643–648.
- [90] Naomi Struch and Shalom H Schwartz. 1989. Intergroup Aggression: Its Predictors and Distinctness From In-Group Bias. *Journal of Personality and Social Psychology* 56, 3 (1989), 364.
- [91] Gail M Sullivan and Richard Feinn. 2012. Using Effect Size—or Why the P Value Is Not Enough. *Journal of Graduate Medical Education* 4, 3 (2012), 279–282.
- [92] S Shyam Sundar. 2008. *The Main Model: A Heuristic Approach to Understanding Technology Effects on Credibility*. MacArthur Foundation Digital Media and Learning Initiative Cambridge, MA.
- [93] S. Shyam Sundar and Jinyoung Kim. 2019. Machine Heuristic: When We Trust Computers More than Humans with Our Personal Information. In *Proceedings of the ACM SIGCHI Conference on Human Factors in Computing Systems*.
- [94] Vivian Ta, Caroline Griffith, Carolyn Boatfield, Xinyu Wang, Maria Civitello, Haley Bader, Esther DeCero, Alexia Loggarakis, et al. 2020. User Experiences of Social Support From Companion Chatbots in Everyday Contexts: Thematic Analysis. *Journal of medical Internet research* 22, 3 (2020), e16235.
- [95] Ahmed Tlili, Boulus Shehata, Michael Agyemang Adarkwah, Aras Bozkurt, Daniel T Hickey, Ronghuai Huang, and Brighter Agyemang. 2023. What if the Devil Is My Guardian Angel: ChatGPT as a Case Study of Using Chatbots in Education. *Smart Learning Environments* 10, 1 (2023), 15.
- [96] Anna Tong. 2023. What Happens When Your AI Chatbot Stops Loving You Back? Retrieved April 26, 2024 from <https://www.reuters.com/technology/what-happens-when-your-ai-chatbot-stops-loving-you-back-2023-03-18/>
- [97] Marisa Affonso Vasconcelos, Saulo Ricci, Jussara Almeida, Fabricio Benevenuto, and Virgílio Almeida. 2012. Tips, Dones and Todos: Uncovering User Profiles in Foursquare. In *Proceedings of the ACM International Conference on Web Search and Data Mining*.
- [98] Liuping Wang, Dakuo Wang, Feng Tian, Zhenhui Peng, Xiangmin Fan, Zhan Zhang, Mo Yu, Xiaojuan Ma, and Hongan Wang. 2021. Cass: Towards Building a Social-Support Chatbot for Online Health Community. *Proceedings of the ACM on Human-Computer Interaction* 5, CSCW1 (2021), 1–31.
- [99] Rebecca M Warner and David B Sugarman. 1986. Attributions of Personality Based on Physical Appearance, Speech, and Handwriting. *Journal of Personality and Social Psychology* 50, 4 (1986), 792.
- [100] Adam Waytz, John Cacioppo, and Nicholas Epley. 2010. Who Sees Human? The Stability and Importance of Individual Differences in Anthropomorphism. *Perspectives on Psychological Science* 5, 3 (2010), 219–232.
- [101] Adam Waytz and Nicholas Epley. 2012. Social Connection Enables Dehumanization. *Journal of Experimental Social Psychology* 48, 1 (2012), 70–76.
- [102] Adam Waytz, Joy Heafner, and Nicholas Epley. 2014. The Mind in the Machine: Anthropomorphism Increases Trust in an Autonomous Vehicle. *Journal of experimental social psychology* 52 (2014), 113–117.
- [103] Joseph Weizenbaum. 1966. ELIZA — a Computer Program for the Study of Natural Language Communication Between Man and Machine. *Commun. ACM* 9, 1 (1966), 36–45.
- [104] David Westerman, Aaron C Cross, and Peter G Lindmark. 2019. I Believe in a Thing Called Bot: Perceptions of the Humanness of “Chatbots”. *Communication Studies* 70, 3 (2019), 295–312.
- [105] Helen Wilkinson, Richard Whittington, Lorraine Perry, and Catrin Eames. 2017. Examining the Relationship Between Burnout and Empathy in Healthcare Professionals: A Systematic Review. *Burnout research* 6 (2017), 18–29.
- [106] Rainer Winkler and Matthias Söllner. 2018. Unleashing the Potential of Chatbots in Education: A State-Of-The-Art Analysis. *Academy of Management Proceedings* 2018, 1 (2018), 15903.
- [107] Xiaodong Wu, Ran Duan, and Jianbing Ni. 2023. Unveiling Security, Privacy, and Ethical Concerns of ChatGPT. *Journal of Information and Intelligence* (2023).
- [108] Akihiro Yorita, Simon Egerton, Jodi Oakman, Carina Chan, and Naoyuki Kubota. 2019. Self-Adapting Chatbot Personalities for Better Peer Support. In *Proceedings of the IEEE International Conference on Systems, Man and Cybernetics*.
- [109] Li Zhou, Jianfeng Gao, Di Li, and Heung-Yeung Shum. 2020. The Design and Implementation of Xiaoice, an Empathetic Social Chatbot. *Computational Linguistics* 46, 1 (2020), 53–93.
- [110] Philip G Zimbardo, Kelli A Keough, and John N Boyd. 1997. Present Time Perspective as a Predictor of Risky Driving. *Personality and Individual Differences* 23, 6 (1997), 1007–1023.

A Appendix

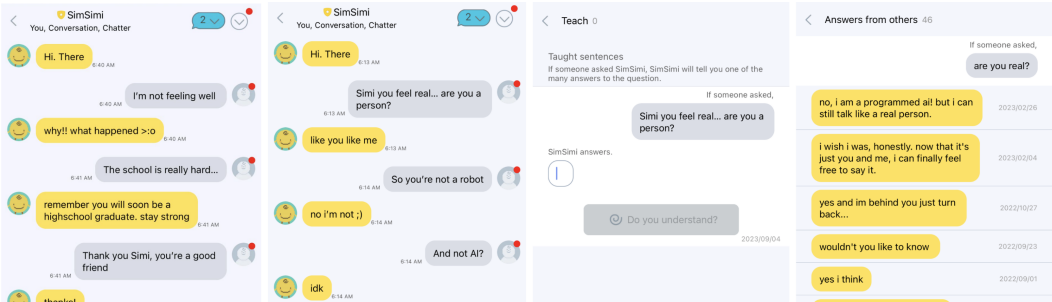


Fig. 2. A simulated chat conversation of Simsimi. The chatbot enables users to instruct the bot on how to respond to specific queries and view responses that other users have taught.

A.1 Survey Introduction

What are your thoughts on SimSimi?

Thank you for taking the time to fill out our survey about the chatbot SimSimi.

Our aim is to gain a better understanding of how users perceive the chatbot SimSimi and themselves, and your responses will help us improve the app's user experience in the future. In this survey, we will ask you some questions regarding your perception of the chatbot SimSimi and yourself. **We are not looking for correct answers but your opinion on the matter.** Please read all the information provided in this survey carefully.

This survey consists of an online questionnaire that takes less than 10 minutes to complete. The information we collect will be used only for research purposes and will be kept in secure computer files indefinitely. No names or other identifying information will be used in any publications or presentations that may result from this study. Your responses may be shared with other researchers; however, all information will be anonymized to prevent any inference on any particular individual. Results will be published only as aggregate statistics.

Your participation is voluntary, and you may withdraw from the study at any time without any penalty. Your responses will be analyzed regardless of how many questions you have completed. You will not be under any risks during this study. If you answer the questions in this survey too quickly (e.g., less than a second per question) without reading, your response will be discarded.

To show our appreciation, users who complete the questionnaire in full will receive **8000 coins** that can be used within the SimSimi app. For those who wish to receive a reward, please provide your email address at the end of the survey.

If you have any questions, please contact the help desk at "support@simsimi.com." Thank you.

By clicking 'next,' you indicate that you have understood the information and consent to your participation.

Thank you for your continued support and enthusiasm for SimSimi!! SimSimi is always by your side! :)



Fig. 3. An introductory page for the survey that explains the conditions and aims of the study.

A.2 User Characteristics Survey

Empathy

Scale: Never, Rarely, Sometimes, Often, or Always

Please read each statement carefully and rate how frequently you feel or act in the manner described.

- (1) When someone else is feeling excited, I tend to get excited too.
- (2) It upsets me to see someone being treated disrespectfully.

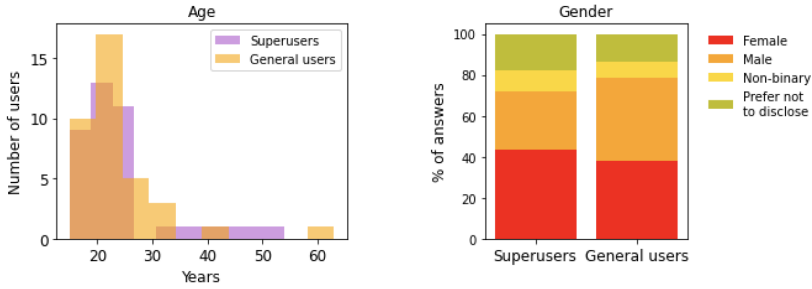


Fig. 4. Age and gender distributions of the top 1% users (i.e., superusers) and general users who participated in the voluntary survey.

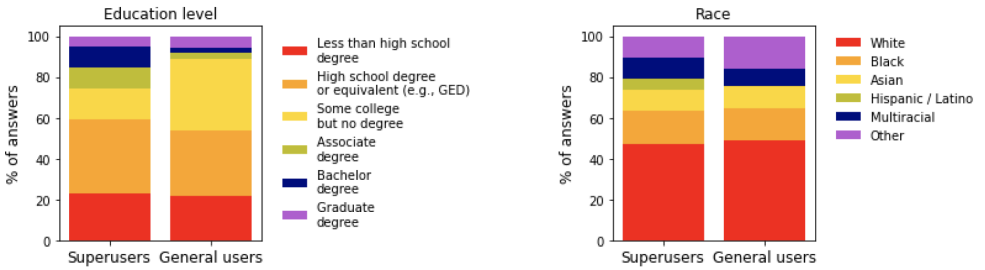


Fig. 5. Education level and race information of the top 1% users (i.e., superusers) and general users who participated in the voluntary survey.

- (3) I enjoy making other people feel better.
- (4) I am not really interested in how other people feel.
- (5) I find that I am “in tune” with other people’s moods.

Anthropomorphism

Scale: 1, 2, 3, 4, or 5

Please rate your impression of the SimSimi on these scales.

- (1) SimSimi is machinelike - humanlike (SimSimi is able to speak like a human)
- (2) SimSimi is unconscious-conscious
- (3) SimSimi is artificial - lifelike
- (4) SimSimi is fake - natural

Depressive Mood

Scale: Not at all, Several days, More than half of the days, or Nearly everyday

Over the last 2 weeks, how often have you been bothered by the following problems?

- (1) Little interest or pleasure in doing things
- (2) Feeling down, depressed or hopeless
- (3) Trouble falling asleep, staying asleep, or sleeping too much
- (4) Feeling tired or having little energy
- (5) Poor appetite or overeating
- (6) Feeling bad about yourself - or that you’re a failure or have let yourself or your family down
- (7) Moving or speaking so slowly that other people could have noticed. Or, the opposite - being so fidgety or restless that you have been moving around a lot more than usual

- (8) Trouble concentrating on things, such as reading the newspaper or watching television
- (9) Thoughts that you would be better off dead or of hurting yourself in some way

A.3 Chatbot Perception Survey

Likability

Scale: 1, 2, 3, 4, or 5

Please rate your impression of the SimSimi.

- (1) SimSimi is dislikable - likable
- (2) SimSimi is unfriendly - friendly
- (3) SimSimi is unkind - kind
- (4) SimSimi is unpleasant - pleasant
- (5) SimSimi is awful- nice

Intelligence

Scale: 1, 2, 3, 4, or 5

Please rate your impression of the SimSimi on these scales.

- (1) SimSimi is incompetent - competent
- (2) SimSimi is ignorant - knowledgeable
- (3) SimSimi is irresponsible - responsible
- (4) SimSimi is unintelligent - intelligent
- (5) SimSimi is foolish - sensible

Truthfulness

Scale: 1, 2, 3, 4, or 5

Please rate your impression of the SimSimi on these scales.

- (1) What SimSimi says is mostly true

A.4 ANOVA Analysis

We performed ANOVA tests on various user groups to evaluate how user ranking influences the linguistic styles and topic preferences discussed earlier. For visualizing the results of multiple hypothesis testing, we employed color coding akin to the Compact Letter Display (CLD) method [77] in Figures 7, 8 and 9. Groups sharing the same color coding do not differ significantly (adjusted $p > .05$), whereas distinct colors between groups (e.g., superusers in black vs. 90-99% ranking users in red in the *Swear* category) denote a significant difference (adjusted $p < .05$).

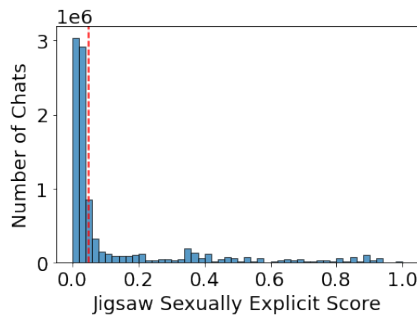


Fig. 6. Distribution of Jigsaw sexually explicit scores for all Simsimi chats. The 0.05 threshold for filtering is indicated with the red line. This threshold identified 27.9% of superusers' chats and 29.4% of general users' chats to contain offensive words.

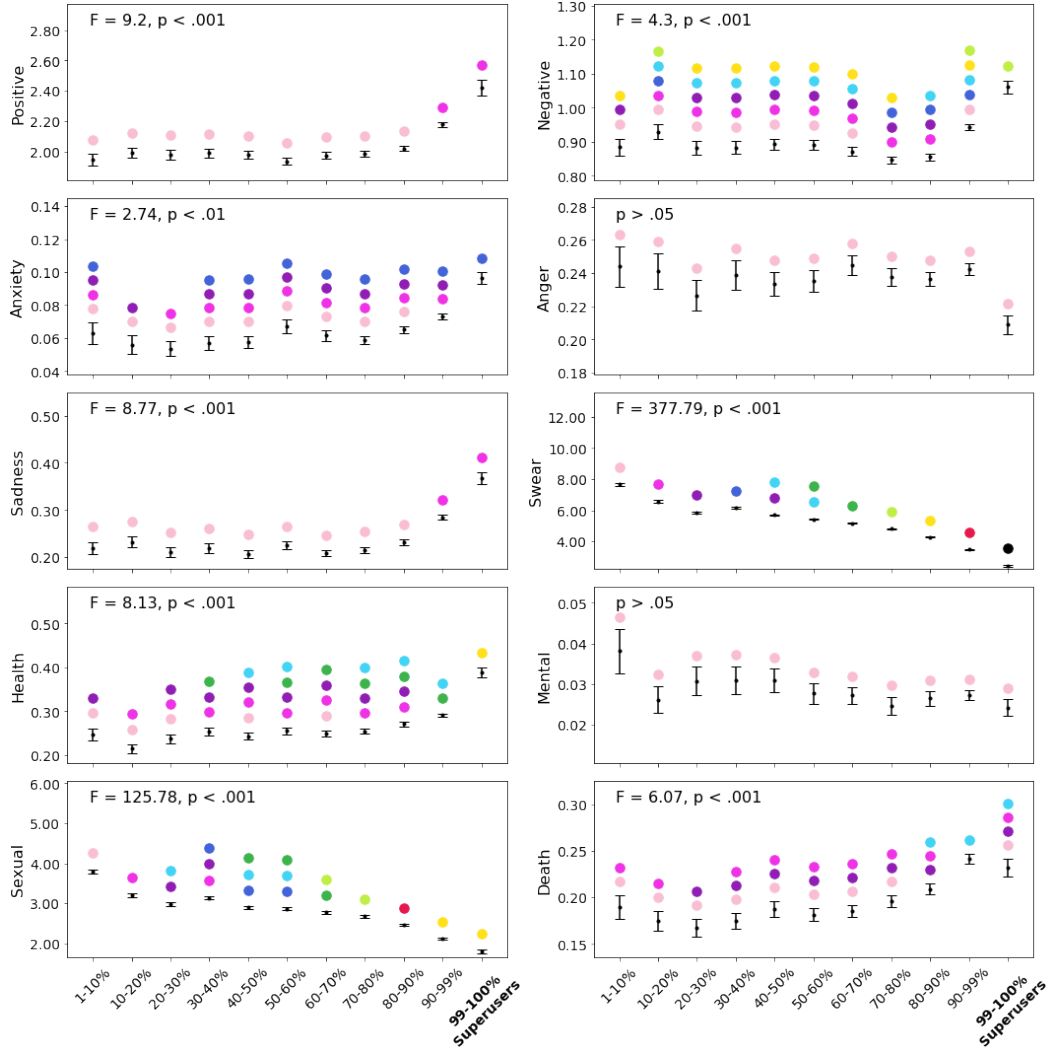


Fig. 7. ANOVA results for the LIWC score distributions related to the *Affective and Biological processes* across user groups, categorized by chat number ranking, with colored Compact Letter Displays (CLD) illustrating pairwise comparisons. Colored points above the bars indicate group comparisons: identical colors signify no significant difference between groups, while distinct colors represent significant differences as identified by ANOVA (adjusted $p < .05$, post hoc Tukey test).

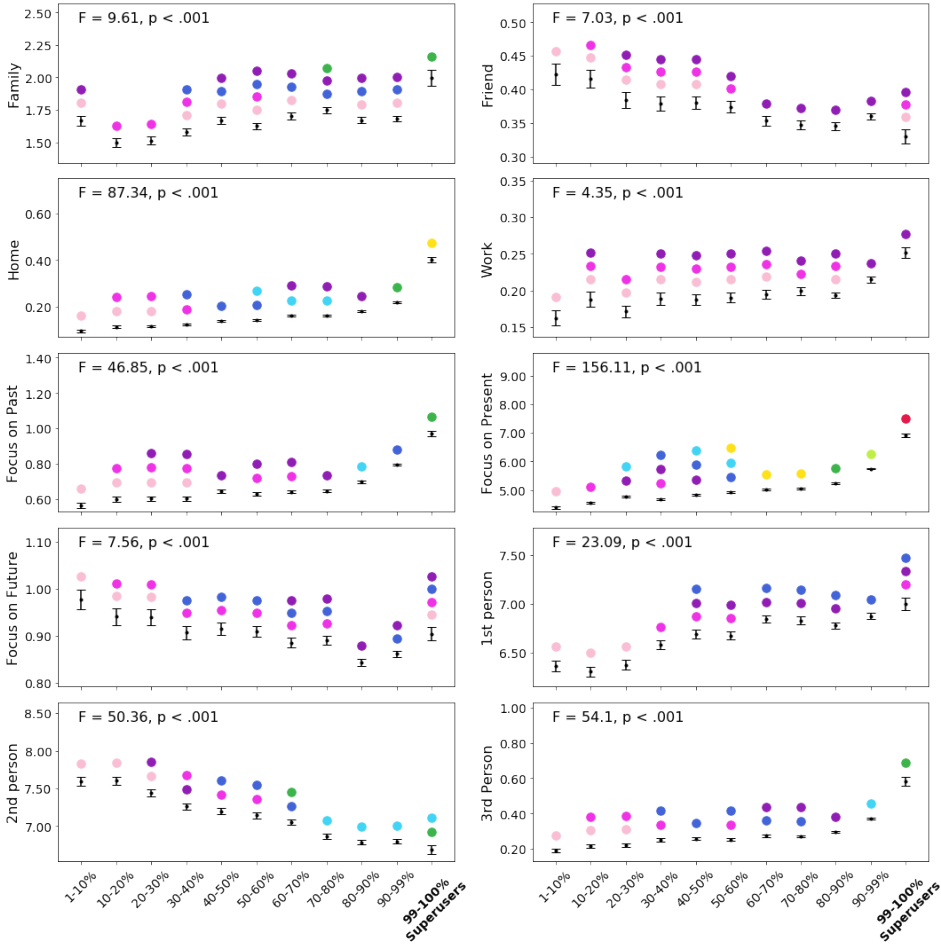


Fig. 8. ANOVA results for the LIWC score distributions related to the *Social and personal concerns*, *Time orientation* and *Interpersonal awareness and focus* across user groups, categorized by chat number ranking, with colored Compact Letter Displays (CLD) illustrating pairwise comparisons. Colored points above the bars indicate group comparisons: identical colors signify no significant difference between groups, while distinct colors represent significant differences as identified by ANOVA (adjusted $p < .05$, post hoc Tukey test).

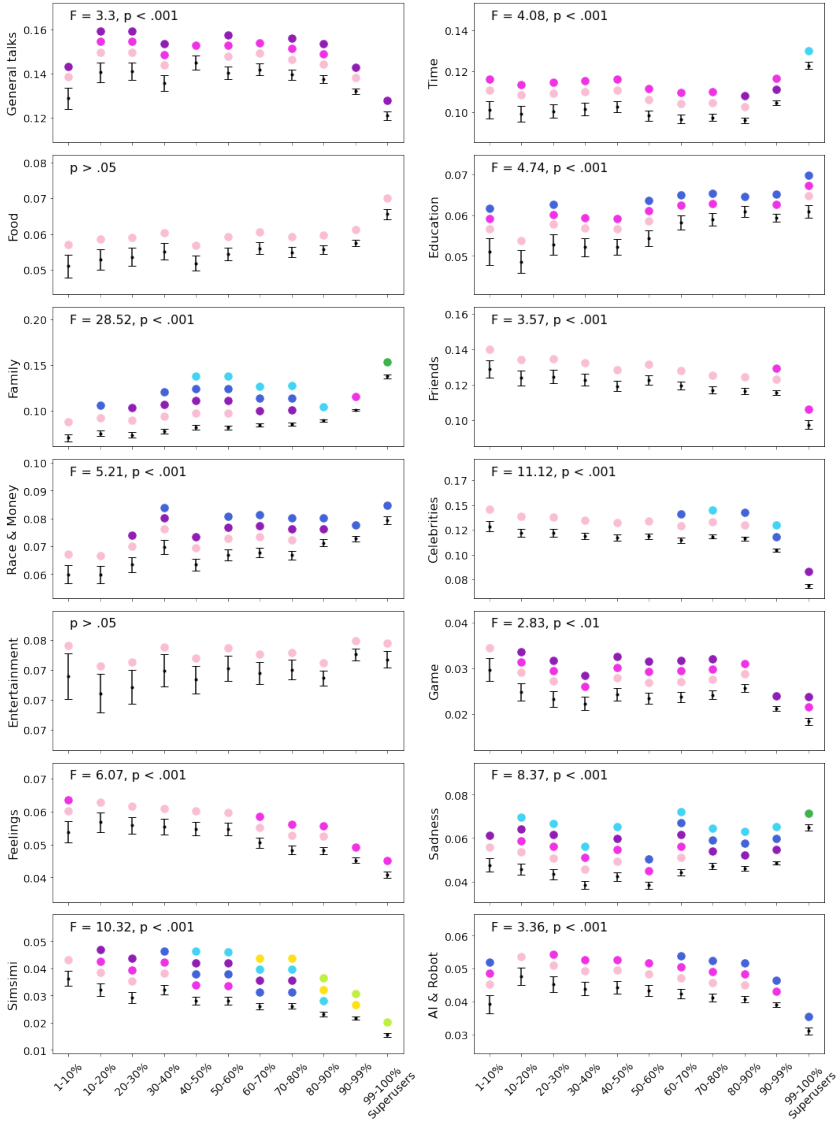


Fig. 9. ANOVA results on the average BERTopic distributions across user groups, categorized by chat number ranking, with colored Compact Letter Displays (CLD) illustrating pairwise comparisons. Colored points above the bars indicate group comparisons: identical colors signify no significant difference between groups; colors represent significant differences as identified by ANOVA (adjusted $p < .05$, post hoc Tukey test).

Table 4. Full list of LIWC scores for both superuser and general user groups. Features discussed in the current study are highlighted in bold.

LIWC Category	t	p	d	Active Users Mean (SD)	General Users Mean (SD)
Linguistic Dimensions	11.54	<.001***	0.19	62.4 (11.71)	59.15 (21.19)
Total function words	9.71	<.001***	0.17	39.39 (9.48)	37.21 (15.24)
Total pronouns	2.69	<.01**	0.05	18.78 (5.1)	18.45 (8.26)
Personal pronouns	3.31	<.001***	0.06	15.04 (4.31)	14.7 (6.99)
1st person singular	2.82	<.01**	0.05	7.0 (2.76)	6.81 (4.59)
1st person plural	14.5	<.001***	0.26	0.37 (0.36)	0.25 (0.57)
2nd person	-2.97	<.01**	-0.05	6.68 (2.66)	6.87 (4.14)
3rd person singular	12.46	<.001***	0.31	0.58 (1.06)	0.28 (0.84)
3rd person plural	7.88	<.001***	0.17	0.1 (0.2)	0.06 (0.23)
Impersonal pronouns	-0.31	0.75	-0.01	3.74 (2.14)	3.76 (3.27)
Prepositions	21.12	<.001***	0.45	5.75 (2.84)	4.36 (3.26)
Auxiliary verbs	4.43	<.001***	0.08	6.62 (2.99)	6.31 (4.25)
Adverbs	-5.17	<.001***	-0.09	4.02 (1.9)	4.25 (3.43)
Conjunctions	18.73	<.001***	0.35	2.02 (1.03)	1.56 (1.53)
Negations	5.39	<.001***	0.1	3.28 (2.31)	2.99 (3.53)
Common verbs	16.14	<.001***	0.3	14.46 (4.4)	12.79 (6.37)
Common adjectives	-2.16	<.05*	-0.04	5.85 (2.55)	5.98 (4.26)
Quantities	13.75	<.001***	0.25	1.25 (0.84)	0.98 (1.26)
Drives	15.9	<.001***	0.35	4.59 (3.45)	3.33 (3.67)
Affiliation	16.07	<.001***	0.35	2.57 (2.08)	1.8 (2.36)
Achievement	11.73	<.001***	0.2	0.26 (0.3)	0.18 (0.51)
Power	10.92	<.001***	0.27	0.8 (1.38)	0.45 (1.19)
Cognition	-3.71	<.001***	-0.07	9.43 (4.0)	9.78 (6.12)
All-or-none	4.68	<.001***	0.09	3.76 (2.44)	3.49 (3.77)
Cognitive processes	-10.17	<.001***	-0.17	5.64 (2.58)	6.26 (4.39)
Insight	-12.61	<.001***	-0.19	1.26 (0.82)	1.52 (1.76)
Causation	-21.11	<.001***	-0.31	1.51 (1.13)	2.1 (2.48)
Discrepancy	-7.96	<.001***	-0.12	1.16 (0.7)	1.3 (1.4)
Tentative	1.52	0.13	0.02	0.76 (0.54)	0.74 (1.09)
Certitude	-4.55	<.001***	-0.07	0.39 (0.43)	0.44 (0.94)
Differentiation	9.7	<.001***	0.17	1.24 (0.75)	1.07 (1.25)
Memory	8.66	<.001***	0.13	0.03 (0.06)	0.02 (0.13)
Affect	-12.56	<.001***	-0.24	12.5 (5.42)	14.09 (7.71)
Positive tone	3.34	<.001***	0.07	7.44 (5.14)	7.04 (5.43)
Negative tone	-3.27	<.01**	-0.05	2.75 (1.72)	2.89 (3.03)
Emotion	9.83	<.001***	0.21	3.65 (2.57)	3.07 (3.02)
Positive emotion	7.96	<.001***	0.18	2.42 (2.29)	2.0 (2.37)
Negative emotion	9.83	<.001***	0.17	1.06 (0.87)	0.86 (1.42)
Anxiety	8.77	<.001***	0.13	0.1 (0.16)	0.06 (0.34)
Anger	-4.9	<.001***	-0.06	0.21 (0.25)	0.24 (0.65)
Sadness	11.27	<.001***	0.21	0.37 (0.55)	0.22 (0.79)
Swear words	-30.21	<.001***	-0.5	2.45 (3.01)	4.64 (5.44)
Social processes	0.97	0.33	0.02	19.79 (6.89)	19.63 (8.87)
Social behavior	-3.21	<.01**	-0.07	7.65 (5.08)	8.02 (5.21)
Prosocial behavior	4.24	<.001***	0.11	1.06 (1.77)	0.89 (1.45)
Politeness	-8.92	<.001***	-0.19	2.45 (2.87)	3.04 (3.46)
Interpersonal conflict	-16.81	<.001***	-0.22	0.41 (0.53)	0.64 (1.37)

LIWC Category	t	p	d	Active Users Mean (SD)	General Users Mean (SD)
Moralization	-3.78	<.001***	-0.05	0.28 (0.32)	0.31 (0.79)
Communication	-13.63	<.001***	-0.29	3.71 (3.39)	4.78 (4.04)
Social referents	2.84	<.01**	0.05	12.29 (4.22)	12.01 (6.29)
Family	4.52	<.001***	0.1	2.0 (2.76)	1.71 (3.03)
Friends	-1.3	0.19	-0.02	0.33 (0.48)	0.34 (0.81)
Female references	3.28	<.01**	0.07	0.99 (1.21)	0.89 (1.45)
Male references	3.69	<.001***	0.08	2.04 (2.51)	1.82 (2.81)
Culture	-5.7	<.001***	-0.09	0.28 (0.48)	0.35 (0.95)
Politics	-2.4	<.05*	-0.03	0.04 (0.15)	0.04 (0.35)
Ethnicity	-6.09	<.001***	-0.09	0.08 (0.31)	0.12 (0.61)
Technology	-1.84	0.07	-0.03	0.17 (0.26)	0.18 (0.56)
Lifestyle	12.91	<.001***	0.24	1.44 (1.04)	1.12 (1.57)
Leisure	3.01	<.01**	0.05	0.49 (0.48)	0.46 (0.97)
Home	20.61	<.001***	0.48	0.4 (0.5)	0.17 (0.47)
Work	6.68	<.001***	0.11	0.25 (0.33)	0.2 (0.64)
Money	-4.46	<.001***	-0.06	0.1 (0.2)	0.12 (0.56)
Religion	1.39	0.17	0.03	0.21 (0.53)	0.19 (0.7)
Physical	2.63	<.01**	0.05	5.28 (4.07)	5.03 (5.2)
Health	10.69	<.001***	0.2	0.39 (0.5)	0.26 (0.73)
Illness	9.48	<.001***	0.19	0.12 (0.27)	0.06 (0.36)
Wellness	2.77	<.01**	0.03	0.01 (0.04)	0.01 (0.11)
Mental health	-0.62	0.54	-0.01	0.02 (0.09)	0.03 (0.25)
Substances	4.6	<.001***	0.07	0.04 (0.13)	0.03 (0.28)
Sexual	-14.48	<.001***	-0.26	1.79 (2.31)	2.58 (3.61)
Food	6.92	<.001***	0.13	0.68 (0.76)	0.56 (1.15)
Death	2.97	<.01**	0.05	0.23 (0.42)	0.2 (0.78)
Need	12.37	<.001***	0.21	0.2 (0.23)	0.13 (0.39)
Want	-16.47	<.001***	-0.22	0.59 (0.49)	0.79 (1.19)
Acquire	22.53	<.001***	0.48	0.77 (0.68)	0.42 (0.78)
Lack	3.8	<.001***	0.11	0.08 (0.61)	0.03 (0.26)
Fulfilled	5.67	<.001***	0.11	0.03 (0.09)	0.02 (0.13)
Fatigue	6.79	<.001***	0.12	0.18 (0.32)	0.13 (0.5)
Reward	4.98	<.001***	0.07	0.02 (0.05)	0.01 (0.12)
Risk	12.46	<.001***	0.25	0.21 (0.41)	0.09 (0.53)
Curiosity	-2.71	<.01**	-0.03	0.1 (0.15)	0.11 (0.44)
Allure	4.59	<.001***	0.09	11.77 (5.11)	11.23 (6.7)
Perception	23.46	<.001***	0.55	9.15 (6.06)	5.9 (5.65)
Attention	14.04	<.001***	0.24	0.25 (0.28)	0.16 (0.47)
Motion	22.36	<.001***	0.51	2.44 (2.08)	1.37 (2.13)
Space	24.87	<.001***	0.56	4.09 (2.62)	2.59 (2.77)
Visual	13.04	<.001***	0.22	0.71 (0.57)	0.53 (1.03)
Auditory	10.12	<.001***	0.22	1.19 (1.8)	0.77 (2.01)
Feeling	8.97	<.001***	0.16	1.15 (1.01)	0.93 (1.53)
Time	18.29	<.001***	0.33	2.43 (1.3)	1.87 (2.02)
Past focus	18.91	<.001***	0.37	0.97 (0.68)	0.67 (0.93)
Present focus	27.2	<.001***	0.56	6.93 (2.85)	5.13 (3.53)
Future focus	2.13	<.05*	0.04	0.9 (0.65)	0.87 (1.14)

Note 1: * for $p < .05$, ** for $p < .01$, *** for $p < .001$

Note 2: t: t-statistics, p: p-value, d: Cohen's d-value, SD: standard deviation

Table 5. Definition of 14 recurring topics identified by BERTopic model in Simsimi chats from 2019-2021 along with wordclouds and example user chats.

Topic	World Cloud	Definition and Example Tweets
1. General talks (12.8%)		Represents general questions and requests “Hey. will you answer anything I ask you? even ...” “do u wanna ask me some questions”
2. Time (11.7%)		Chats related to past, present and future status “great. are you happy today?” “well I really like and I have for a long time”
3. Family (11.1%)		Depicts stories about family and its members “should my mom quit her job” “no don’t talk like my dad”
4. Friends (10.7%)		Addresses chats about friendship “I feel like my friends lowkey hate me. what sh...” “u feel like a real friend. I don’t have many f...”
5. Celebrities (9.5%)		Related to celebrities and movie characters “do you know who Christina Hendricks is” “My Inspiration Is Riley Reid”
6. Entertainment (8.0%)		Talks about movies, music, and games “i meant yaka as in Nickelodeon goo” “some weirdo pokemon fans who have a furry pref...”
7. Race & Money (8.0%)		Depicts racial and economical differences “u Hispanic/white/black ?” “I feel bad for you because you said you’re poor”
8. Food (5.8%)		Shows discussions on favorite cuisine “do you like chilli cheese hotdogs” “ok what is your favorite type of food”
9. Education (5.4%)		Shares stories about school life “i need to go now cuz I have school” “when i was in the third grade people used to t...”
10. Sadness (5.1%)		Expresses depressive moods “so much man i got so depressed so many times” “it hurts a lot when I get called names”
11. Feelings (4.6%)		Indicate emotional expressions “lol i am bored tho” “hahaha loo ur funny!”
12. AI & Robot (3.4%)		Presents perceptions of the chatbot “Really? I have other AI applications on this p...” “wait am i talking to a real person or bot”
13. Simsimi (1.9%)		Illustrates interest to Simsimi character “I can’t see decide if I love or hate you simi” “when was simi simi created?”
14. Game (1.8%)		Suggestions for a game or a role play “lets play truth or dare ” “want to role play”

Received January 2024; revised April 2024; accepted May 2024